DOCUMENT 00 · MASTER ENGINEERING BASELINE (MODULE)

MCRS

Modular Captured Ride System · CMFSB

Captured Mechanical Floating Safety Body

Always captured. Soft when you want comfort. Locked when you need control. Settles on passive support when power is gone.

Field	Controlled value
Document ID	MCRS-DOC-00-R8.09 (Master)
Issue date	2026-10-12
Founder / Legal Owner	Amirbahador Atai (individual rights holder; assignment to a legal entity is a separate controlled act)
Licensable module	MCRS-MOD-001 (four- or six-tower CMFSB isolation module)
Classification	T1 Confidential / NDA controlled
Document class	Host-neutral Master (module values). Host baselines MCRS-DOC-00-R5.40 (hybrid) and MCRS-FE-DOC-00-R7.09 (electric) are subordinate for module values.
Supersedes	MCRS-DOC-00-R8.08. Host baselines R5.40 and R7.09 are issued with this Master and supersede R5.39 and R7.08. Review RX-01 to RX-06 (R8.09) applied: gate criteria linked to REQ-M30 and REQ-M32 to REQ-M36, assumption and owner-decision register, release-readiness and contract-use statement, release record (Appendix N).
Status	Proposed baseline, pending owner release and independent technical review. Evidence level L0, TRL 2–3.
Independent Technical Reviewer	To be appointed no later than the earlier of 2026-12-05 and G1 closure, and before any external safety-performance credit.
Approval authority	TBD after independent review

This document is a design baseline and diligence instrument. It does not certify range, crashworthiness, homologation, ASIL, production readiness, patentability or freedom to operate. OPEN items remain OPEN until evidence closes them. Module values are controlled only by this Master. Host values are controlled only by the host baselines.

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1. Purpose, scope and positioning

This baseline defines the licensable CMFSB isolation module (MCRS-MOD-001) independently of any host powertrain. An electric and a hybrid reference host exist only to study interface, mass, packaging and energy; their values live in their own baselines.

Document	Role	Governs
MCRS-DOC-00-R8.09 (this file)	Host-neutral Master	Module values, safety invariants, modes, requirements
MCRS-DOC-00-R5.40	Hybrid host baseline	Hybrid-host values only
MCRS-FE-DOC-00-R7.09	Electric host baseline	Electric-host values only

> One value, one place. Module values are controlled only in §7 and Part B. Host values are controlled only in the host baselines. A value may appear a second time only in a section labelled citation or mirror (§9, Annexes E and H, Appendices I and K); the controlling source prevails on any difference and Appendix D checks the match.

1.1 Value statement

Occupant compartments in specialty, defence-utility, ambulance and limited-series vehicles receive road vibration that degrades comfort, task performance and, where relevant, clinical stability. Soft isolation alone can reduce handling stiffness in rapid manoeuvres. CMFSB addresses both: the upper body stays mechanically captured at all times, with a COMFORT isolation envelope, a SPORT mechanical lock and passive support on power loss. Reference application RA-1 (ambulance / specialty medical transport) is the context for comfort requirements; no clinical outcome is claimed.

State	Experience	Architecture behaviour	ID
COMFORT	Road harshness isolated from the cabin	Locks released; macro and micro stages active; leveling active	S2
SPORT	Body feels planted in rapid manoeuvres	All required locks confirmed; isolation stages held	S2S
Power loss	Cabin settles on a known mechanical path	Powered valves fail-closed; passive support; no software required	S4

1.2 What is offered

- Path A (primary): licence of the module under a paid Option (SIM-001 run and ICD-001 draft); the buyer may bring its own host.
- Path B: host engagement studies using the electric or hybrid reference host; no vehicle sale is implied.
- Path C: IP-transaction discussion under counsel (T3 only).

1.3 Document control

The host baselines MCRS-DOC-00-R5.40 (hybrid) and MCRS-FE-DOC-00-R7.09 (electric) are issued together with this Master on 2026-10-12 and supersede R5.39 and R7.08. Any conflict on module values is resolved in favour of this Master. Revision identities are unique per content: no two different documents share an ID.

2. Executive summary

The CMFSB module places a captured, isolated upper safety body on four or six capture towers. Each tower combines a hydraulic macro stage, a force-bounded Lorentz micro stage, progressive stops and a SPORT positive lock. Safety rests on mechanical capture, passive support and stops, never on the actuator or on software.

Module item	Value	Status
Towers per host	4 (Class S); 4 or 6 (Class L)	BASELINE
Macro / micro stroke	±50 mm (100 mm total) / ±7.5 mm	TARGET
Force peak / RMS per tower	±2.5 kN / ≈0.7 kN	STUDY
Km	≥63 N/√W at 2–5 mm gap	ASSUMPTION
Core tower mass	P50 48 kg (target ≤55); P90 ≈59 kg RSS / 67 kg arithmetic; limit until G2 = 67 kg	TARGET / OPEN
Corner mass planning	Class S ≈70 kg · M ≈85 kg · L ≈95 kg	STUDY
Control bandwidth / first mode	≥20 Hz / ≥60 Hz	TARGET
Mode time / life	≤3 s / ≥10,000 h equivalent	TARGET
Lateral-g interlock	Inhibit above 0.35 g; re-enable below 0.30 g	TARGET (temporary)
Module supply	48 V class, galvanically isolated from host HV	STUDY
Evidence level	L0 architecture; TRL 2–3	BASELINE

2.1 Readiness assessment (module level)

Dimension	Position	Principal exposure	Closure
Architecture	Defined and consistent at L0	No hardware or simulation evidence	SIM-001 (G1), tower bench (G2)
Actuator	Km is an assumption	Thermal and force-gap limits unproven	ACT-001 (G2)
Safety	Invariants and screening FMEA	No HARA / FTA / independence argument	SAF-001 (G2)
Packaging	Re-baselined heights 1,950 / 2,130 mm; stack residual 0 / +10 mm	Closure depends on 170 mm of unvalidated levers; zero margin on V-L	G1 3-D datum stack
Mass	P50 within target; P90 open	Class S 70 kg corner has no design evidence; V-H four-tower P90 427–565 kg (six-tower option 605–815 kg)	G1 Class S study, G2 supplier data
IP	Ownership chain and filing OPEN	No filing receipt on record	Counsel, provisional filing
Commercial	Path A Option structure defined	No validated interface set	ICD-001 draft (G1)

3. Product definition and architecture

The host is defined around the CMFSB load and isolation architecture; it is not a powertrain retrofit. The MCRS tower set is the sole mechanical connection between the two body pieces in normal operation. Active isolation works inside a bounded mechanical safety envelope; comfort control, handling stiffness and crash containment are coordinated but independently bounded.

3.1 Layer stack (top to bottom)

Layer	Function	Safety role
Upper safety cell	Occupant and mission body (CFRP, roll structure, seats, closures)	Captured at all times
Micro stage (Lorentz)	High-bandwidth isolation within ±7.5 mm	Force-bounded; never a crash or horizontal path
Macro stage (hydraulic)	Leveling, ±50 mm body-relative travel, lock seats	Maintains geometry and ride envelope
Progressive stops	Energy absorption on abnormal vertical events	Primary abnormal vertical energy path
SPORT positive lock	Pin, claw or hydraulic clamp at the seat band	Handling stiffness and load sharing; not a replacement for stops
Capture frame and links	Longitudinal, lateral and yaw restraint	Primary horizontal path
Lower chassis hardpoints	Load transfer into the host platform	Structural continuity

Figure 1 (conceptual). Two-piece body: the upper cell is captured on MCRS towers above the lower host chassis. V-L: four towers (Class S). V-H: four towers on both hosts (owner decision D-02); six towers remain an option of MCRS-MOD-001. Vertical support is passive macro support; horizontal loads go through the capture frame and links; abnormal vertical energy goes through progressive stops.

3.2 Independent load and force paths

Load / function	Primary path	Secondary / containment
Static vertical support	Passive hydro-pneumatic macro support	Progressive stops / chassis structure
COMFORT dynamic force	Lorentz micro actuator (force-bounded)	Macro stage + passive support
Large displacement, leveling	Hydraulic macro stage	Stops
SPORT stiffness	Positive locks + seat faces	Capture frame
Longitudinal / lateral / yaw	Capture frame, links, horizontal stops	SPORT lock faces
Crash / abnormal vertical	Progressive stops + structural hardpoints	Capture frame / links
Power loss	Passive hydraulic / compliant support (S4)	Mechanical capture
Battery crash load	Lower chassis + protected HV enclosure	Independent containment structures

The Lorentz stage is an actuator option. If another actuator family replaces it, the safety invariants and interface requirements still apply under configuration control. Hold is an optional, rate-limited secondary restraint (orifice and/or friction at the macro stage); it is distinct from the SPORT lock and the stops and is not a primary crash path.

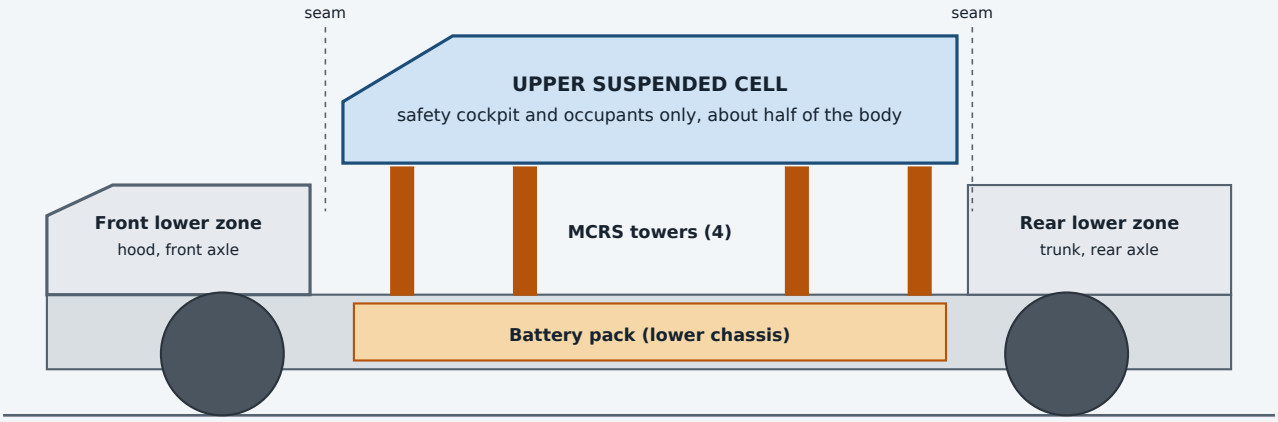
3.3 Qualitative comparison

Approach	Comfort	Handling stiffness	Positive lock	Independent capture	Power loss
Soft air / hydro-pneumatic	Good	Reduced in rapid manoeuvres	No	Limited to suspension travel and bump stops	Settles on bump stops
Adaptive dampers	Improved trade-off	Improved	No	No	Fixed or fail-soft damping
CMFSB / MCRS (target)	Targeted (unproven at L0)	Targeted via SPORT lock	Yes (SPORT)	Yes: stops, frame, links	Settles on passive support, captured

Qualitative only; no measured superiority is claimed at evidence level L0.

3.4 Body zoning and load boundary (owner decisions D-05, D-07)

Only the safety cockpit is suspended: the passenger cell with the occupants' space, about half of the overall body (planning ≈2,500 mm of the 5,000 mm V-L body and ≈2,750 mm of the 5,500 mm V-H body). The hood, front wheel housings, trunk, rear body and wheel housings are attached to the lower body, and the lower chassis forms a cradle around the cell. The battery, e-drive or ICE, fuel tank, axles, brakes, steering and the trunk payload are all on the lower chassis.



Battery, powertrain, fuel, hood and trunk payload weigh on the lower chassis only. The module carries only the cockpit.

Figure 1b (conceptual, not to scale). Side view of the zoning: the upper cell rides on the MCRS towers; battery, powertrain and axles belong to the lower chassis.

Zone	Contents	Carried by	Loads the module?
1 Upper suspended cell (safety cockpit)	CFRP safety cell, roll structure, seats, closures, occupants (no payload, no battery)	MCRS towers	Yes: this is the upper sprung mass
2 Front lower zone	Hood, front wheel housings, front axle, front crash structure	Lower chassis	No
3 Rear lower zone	Trunk, rear body and wheel housings, rear axle(s), payload	Lower chassis	No
4 Lower chassis	Battery pack (electric V-L 275 kWh / V-H 300 kWh; hybrid 20 / 40 kWh), e-drive / ICE, fuel tank, brakes, steering, thermal system	Wheels and suspension	No: never loads the module

How the battery relates to the upper level. It adds no load to the module or the cell. It matters only through (a) total vehicle weight against the GVW ceiling and (b) its thickness inside the lower chassis, which sets the chassis height in the packaging stack (pack volume ≈1,020–1,665 litres). The tower depth shared with the chassis and battery floor (60 mm lever) is the only place where battery packaging and module packaging overlap.

Seams. The joints between the moving cell and the fixed front and rear bodywork shall accommodate ±57.5 mm of normal relative motion (50 mm macro + 7.5 mm micro) plus the stop stroke (50 mm placeholder) without contact, using flexible seals or bellows; values are frozen in ICD-001. Seam water, dust, noise and pinch protection is not yet designed (OPEN).

Payload location (decision D-07). Payload is carried in the trunk on the lower body and does not load the module; the upper sprung mass is the cockpit structure plus occupants (and the optional protection package where fitted). Had payload been carried in the cabin, the upper mass would rise by 100 kg (V-L) and 705 kg (V-H electric), and the V-H electric 20 g corner load from ≈101 to ≈135 kN. Trunk payload loads the lower chassis and rear axle instead (STR-001, G1).

4. Safety invariants and safety layers

§4 is the controlling statement of the safety invariants. Appendix I and both host baselines mirror it; the mirrors are labelled and §4 prevails on any difference.

Invariant	Rule
Mechanical capture	No operating state permits loss of geometric capture; no free levitation.
Static support	Vehicle weight on passive or fail-safe compliant paths, never the sole magnetic path.
Power loss (S4)	Passive support; powered leveling-supply and hold valves fail-closed; the hydro-pneumatic spring path (cylinder to accumulator) stays connected; software aliveness not required.
Crash path	Progressive stops + structure, independent of closed-loop magnetic force.
Mode transition	SPORT↔COMFORT never enters unconstrained free-float.
SPORT lock fault	Lock failure shall not defeat capture or progressive stops; SPORT is never declared with an incomplete lock set.
Horizontal restraint	Longitudinal / lateral / yaw loads resolved by dedicated paths and/or SPORT lock.
Override limit	Driver override cannot defeat the lateral-g interlock or a critical fault state.
Manual rescue	Two-step rescue available under power loss and hydraulic failure.
HV isolation	Traction HV galvanically isolated from the module control domain; a single HV fault shall not defeat capture or S4.
Thermal / fire	Battery runaway containment is a lower-chassis responsibility and shall not propagate into upper-body capture integrity.
Single-path rule	No single active-control path is credited as the primary crash path.
Lock state on power loss	Locks remain engaged only if already engaged at the instant of loss (SPORT). From COMFORT the locks stay released. Logic frozen in ICD-001 / LCK-001 (REQ-M17 / M20).

4.1 Layered safety response

Safety layer	Failure addressed	Design response
Mechanical capture	Link / carriage failure	Capture frame with redundant restraint paths
Passive support	Power loss	Hydro-pneumatic / compliant support, fail-safe valves
Progressive stops	High-energy vertical event	Defined force–displacement curve (STR-001)
Horizontal restraint	Braking, cornering, crash	Dedicated links, stops and lock faces
Positive SPORT locks	Mode stiffness, lock integrity	Per-lock confirmation; incomplete set inhibits SPORT
HV safety	Electrical crash, isolation fault	HVIL, isolation monitor, service disconnect, crash inhibit
Thermal safety	Battery / inverter overheating	Redundant sensing, derating, propagation-resistant pack, vent routing
Control safety	Bad command, software fault	Bounds, diagnostics, degraded state, independent monitor

No ASIL assignment is claimed. High-severity (S ≥ 9) items require explicit pre-G2 review independent of RPN score.

5. Control system, modes and state machine

5.1 Controller partition

Domain	Responsibility
Vehicle domain controller	Driver intent, drive state, dynamics, high-level requests; cannot bypass MCRS interlocks
MCRS Body Control ECU (BCE)	Mode arbitration, corner coordination, command bounds, lock sequencing, leveling, diagnostics
Independent safety monitor	Force / stroke / mode bounds, critical interlocks, watchdog and plausibility, where adopted
Corner controllers	Local sensing and actuator current / valve commands, bounded by BCE
BMS	Battery state, HV safety, contactor control, thermal limits
Brake controller	Primary service braking and regenerative coordination

Comfort control may be sophisticated; safety bounds remain simple, deterministic and independently monitorable. The monitor may be called independent only after power, clock, processor, software, communication and sensor dependencies are analysed and tested. OTA or calibration changes shall not bypass configuration control.

5.2 State machine

State	Name	Rule
S0	Safe / captured / power-off	No powered release; passive capture retained (equivalent to S4)
S1	Precharge / self-test	Capture healthy; auxiliary power path checked
S2	COMFORT	Macro + micro isolation active; capture mandatory
S2S	SPORT	Seat band; all required positive locks confirmed; stages held
S3	Degraded	Reduced authority; force and travel bounded
S4	Power-loss capture	Passive support; fail-safe valves; no software dependency for retention
S5	Impact preparation	High-rate unlock inhibited; stops prepared
S6	Impact / arrest	Progressive stops active; no actuator authority required
S7	Post-event safe	No automatic re-enable; manual clearance required
S8	Maintenance lockout	Physical lockout; stationary only

Any powered state goes to S4 on power loss (engaged locks stay engaged only if already locked; from COMFORT they stay released). S4 exits only to S1 after power restore and self-test. S2↔S2S only under the lateral-g policy with complete lock confirmation. The full transition table, including S5–S8 exit conditions, is required in ICD-001. Detection of power loss ($t_{det,max}$) must be hardware/passive so that S4 without software is not contradicted.

5.3 Mode sequences

Transition	Sequence
COMFORT → SPORT	Request accepted (no inhibit, lateral-g met, no fault) → macro drives to seat band → seat contact at required corners → locks engage → all required locks confirmed (incomplete set aborts to COMFORT) → stages held → S2S
SPORT → COMFORT	Allowed only under lateral-g policy → lock release commanded → all report released → macro envelope re-enabled → micro control re-enabled → S2
Power loss from COMFORT	Immediate S4; active force removed; passive support retained; locks stay released
Power loss from SPORT	S4 / fail-locked; locks remain engaged until controlled recovery
Critical fault	S3 / S4 as applicable; no automatic comfort restoration

6. Power-loss behaviour and lateral-g policy

6.1 REQ-M07 power-loss structure

After loss of commanded active power the system shall: (1) detect the loss within t_det,max (hardware/passive path); (2) enter S4; (3) close the relevant valves within t_valve,max; (4) remain inside the capture envelope; (5) stay below z_max and v_max; (6) prevent uncontrolled free-float; (7) settle in the defined S4 hydraulic state with passive support at or above the class static floor, with residual capacity and environment stated, under electrical loss and the defined hydraulic failure cases. State definition alone is not acceptance.

Limit to freeze	Where	Verified by
t_det,max, t_valve,max	ICD-001 (G2)	Bench + fault injection (L2/L3)
z_max, v_max, settle limit	ICD-001 (G2)	Single-tower power-cut test
Residual stiffness and damping in S4	ICD-001 (G1/G2)	L2 bench
Static support floor after hose-burst case	STR-001 (G2)	Hydraulic failure test
Energy reserve for safe-state functions	ENE-E-001 / ENE-H-001	Power-loss timing test

6.2 Lateral-g policy (REQ-M21)

Element	Rule
Inhibit	SPORT→COMFORT inhibited immediately when |ay| > 0.35 g, for automatic and manual requests
Re-enable	Only after |ay| stays below 0.30 g for the hold time
Driver override	May queue a request; cannot defeat the interlock
Filter and latency	Validated filter and measurement-to-inhibit latency budget, both biased to inhibit
Sensor faults	Validity, plausibility, disagreement thresholds, stale-data handling; a single lateral sensor cannot alone trigger an unsafe transition

Thresholds are temporary TARGET values; SIM-001 and HIL-001 freeze final thresholds, hysteresis, filtering and latency. PARAM-M21-ROLL-GRAD is defined at G2.

7. Module mass, hydraulics and parameters

7.1 Corner assembly mass (Class L planning)

Element	P50 kg	P90 kg	Deviation (P90 − P50)	Scope
Local structure (per tower)	18	25	7	Core
Lorentz + local carriage	22	30	8	Core
Sensors + local electronics	8	12	4	Core
Core tower subtotal	48	67	RSS 11.4	Target ≤55 kg P50 met by estimate; limit until G2 = 67 kg
Cylinder + accumulator share	15	22	7	Corner
Valves / hoses / seals	5	8	3	Corner
Progressive stops	8	12	4	Corner
SPORT lock share	6	10	4	Corner
Harness / dampers / bellows	4	6	2	Corner
Pump + reservoir share (¼)	8	12	4 (16 for the shared pump)	Shared
Full corner assembly (Class L)	94	137		STUDY; ≈95 kg in host WBS

Core-tower RSS is $48 + \sqrt{(7^2 + 8^2 + 4^2)} = 59.4$ kg. The 67 kg arithmetic value stays the limit until G2 supplier correlation data; a 60 kg limit is accepted only with correlation evidence.

7.2 Module mass decomposition and P90 method

Authoritative decomposition for host mass tables: dedicated corner 86 / 125 kg (P50 / P90), shared pump 32 / 48 kg, distribution 17 kg (no P90 deviation assumed). Four-tower Class L P50 = 4 × 86 + 32 + 17 = 393 kg; six-tower P50 = 6 × 86 + 32 + 17 = 565 kg (one shared pump in both).

P90 method. Per-element deviations of the dedicated corner (7, 8, 4, 7, 3, 4, 4, 2 kg) combine to 14.9 kg RSS per corner. Corners are assumed independent: $\sqrt{n} \times 14.9$. The shared pump deviation (16 kg) is one further independent term in the same quadrature sum, so the deviation is $\sqrt{(n \times 223 + 16^2)}$. It is not added linearly.

Case	4-tower Class L	6-tower Class L	Meaning
P50	393 kg	565 kg	Budget value
P90, RSS (pump independent)	427 kg (+34)	605 kg (+40)	Method above; independence assumed
P90, RSS (pump fully correlated, linear)	439 kg (+46)	618 kg (+53)	Sensitivity; sensible upper RSS bound
P90, arithmetic (fully correlated)	565 kg (+172)	815 kg (+250)	Worst case: 4 × 125 + 48 + 17; 6 × 125 + 48 + 17

Reserve coverage: hybrid V-H reserve 150 kg covers both RSS cases but not the arithmetic case (+172 kg). Electric V-H (four towers, module 393 kg) reserve 220 kg covers all three cases, leaving 48 kg in the arithmetic case; the reserve is a programme reserve shared with every other group. Corner independence is itself an assumption until G2 supplier correlation data. Six towers (a module option, not baselined on either host) add ≈172 kg over four towers (565 − 393); the 605 / 618 / 815 kg figures apply to that option only.

7.3 Per-class corner mass and static support floor

Class	Mapped use	Corner P50	Module P50	Static basis per corner	Floor (1.25 × Fmax)
S	Light host (four seats)	≈70 kg (target)	≈280 kg (4 × 70, corner-only; distribution allocation OPEN)	Occupied share 310–350 kg	≥4.5 kN
M	Intermediate	≈85 kg	≈340 kg	Share 450–550 kg	≥7.0 kN
L	Heavy host	≈95 kg (94 incl. pump share)	393 kg (4 towers) / 565 kg (6 towers), incl. 17 kg distribution	Reference max 787.5 kg	≥10.0 kN

Class S exposure. The 70 kg corner has no design evidence. If a Class S corner equals the Class L architecture (94 kg), the four-tower module grows by 96 kg (corner-only) to 113 kg (with the 17 kg distribution). The hybrid V-L reserve (113 kg after D-08) and the electric V-L reserve (140 kg) both cover it; with the 275 kWh pack the electric V-L design GVW stays 43 kg inside the 4,200 kg ceiling. With payload in the trunk the V-L occupied share per corner is ≈295 kg ((780 + 400) ÷ 4), below the 310–350 kg basis; the 4.5 kN floor is kept as the conservative basis. OPEN: G1 Class S study (REQ-M30).

8. Module electrical and auxiliary energy

Module supply is 48 V class, galvanically isolated from host HV. Peak coil power per tower is $(F \div Km)^2 = (2{,}500\ N \div 63\ N/\sqrt{W})^2 \approx 1.575$ kW, so the micro-stage array peaks at ≈6.3 kW (four towers) and ≈9.45 kW (six towers). These are peak, not RMS, values and exceed the baseline DC-DC ratings of 3 kW and 4 kW. A local energy buffer (48 V battery or supercapacitor) is therefore mandatory for both configurations. The buffer serves comfort performance only; the S4 safety case does not rely on it. Safe-state energy reserve is verified under ENE-E-001 (electric) and ENE-H-001 (hybrid); power-loss detection and valve-closure timing are frozen in ICD-001.

Domain	Voltage	Isolation	Function
Module control / Lorentz	48 V class	Galvanic from host HV	Micro actuators, sensors, BCE
Hydraulic valves / hold	48 V / 12 V	Fail-closed on loss	Leveling, hold, lock actuation
Host traction HV	High voltage	Isolated from module	e-drive, HV battery (host responsibility)
Local energy buffer	48 V	Module domain	Peak shaving for Lorentz; mandatory in both configurations

9. Host interface envelope and host comparison

The module interface envelope (hardpoints, hydraulic ports, connectors, space claim) is defined in ICD-001. Host geometry, mass and energy are controlled by the host baselines and are summarised here for comparison only.

9.1 Packaging re-baseline (owner decision D-03)

SPORT height is re-baselined to 1,950 mm (V-L) and 2,130 mm (V-H) at unchanged ground clearance (200 / 300 mm). The 170 mm closure levers are adopted as the stack allocation (tower depth shared with chassis, travel + free play 110 mm, H-point to roof 960 mm, floor-to-H-point 220 mm, thinner shell) and a 50 mm placeholder covers stop stroke and micro-stage clearance until STR-001. The stack residual is 0 mm (V-L) and +10 mm (V-H).

> Conditional closure. The residual depends on levers that no study has validated, including free play reduced from 40 to 10 mm and 60 mm of shared tower depth. V-L has zero margin. Status stays OPEN — P0 until the G1 3-D datum stack confirms it. Mass and CdA sensitivity to the +250 / +130 mm height is open: the host range screens scale CdA with height (0.99 / 1.60 m²) and assume no body-mass change.

9.2 Reference host comparison (citation of R5.40 and R7.09)

Item	V-L electric	V-L hybrid	V-H electric	V-H hybrid
Powertrain	BEV dual e-axle	ICE V8 4.0 L TT + e	BEV 6×6, three e-axles	ICE V12 6.0 L TT + e
Peak power	≈350–380 kW	ICE ≈430–450 + e 150–180 kW	≈480–550 kW	ICE ≈460–500 + e 200–250 kW
On-board energy	275 kWh usable	80 L + 20 kWh	300 kWh usable	150 L + 40 kWh
Range target (STUDY)	≥800 km	1,000 km (cycle ≤70 km/h avg)	≥800 km utility (≤≈70 km/h)	700–800 km
Overall height (SPORT)	1,950 mm	1,950 mm	2,130 mm	2,130 mm
Towers	4 (Class S)	4 (Class S)	4 (Class L)	4 (Class L)
Curb target	3,544 kg	2,778 kg	5,695 kg	4,983 kg
Design GVW	4,044 kg	3,278 kg	7,000 kg (payload residual 705 kg)	5,983 kg
GVW ceiling	4,200 kg	4,200 kg	7,000 kg	7,000 kg
Margin to ceiling	156 kg	922 kg	0 kg	1,017 kg
Stack residual	0 mm (conditional)	0 mm (conditional)	+10 mm (conditional)	+10 mm (conditional)
Evidence	L0	L0	L0	L0

Owner decisions of 2026-10-12. D-01: the V-L engagement ceiling is 4,200 kg for both V-L hosts (programme policy change under Appendix B). D-06: V-L electric uses a 275 kWh usable pack (V-H electric keeps 300 kWh), giving a margin of +156 kg, or +43 kg even with the Class S module at 393 kg; at 200 Wh/kg pack specific energy the V-L design is 150 kg over the ceiling (sizing rule: Appendix M.7). D-08: the hybrid V-L reserve is raised from 70 to 113 kg so that it covers the Class S exposure (curb 2,778 kg, design GVW 3,278 kg, margin 922 kg). The electric V-L design GVW (4,044 kg) is above the 3,500 kg threshold; the hybrid V-L design GVW (3,278 kg) is below it and only the 4,200 kg ceiling is above it (REG-001). D-02: V-H uses four Class L towers on both hosts; six towers remain an option of MCRS-MOD-001. D-07: payload rides in the trunk, so the 20 g corner load is ≈99 kN (hybrid) and ≈101 kN (electric). D-03: heights as in §9.1.

10. Structural principles

The module carries only the upper sprung mass (Zone 1 in §3.4); battery, e-drive, ICE, fuel and trunk payload sit on the lower chassis and never load it. Structural screening loads are based on upper sprung mass, not GVW. Corner loads assumed equal share (÷4 or ÷6) are a magnitude screen only. STR-001 must calculate real corner distribution, including CG location, occupant and payload placement and load transfer under 1 g braking or cornering. Progressive stops provide the primary abnormal vertical energy path with a defined force–displacement curve (STR-001). Capture frame and links provide the primary horizontal path. SPORT locks are not a substitute for stops. The packaging stack carries a 50 mm stop-stroke and clearance placeholder, which Appendix M.5 ties to a design vertical impact speed (OPEN, REQ-M33).

11. Requirements (selected module)

ID	Statement (summary)	Status
REQ-M07	Power-loss structure (detection, valves, envelope, settle)	OPEN — ICD-001
REQ-M11	Module interface and ICD compliance	OPEN — ICD-001
REQ-M16	Class L sizing basis and corner mass targets	STUDY
REQ-M17/M20	Lock state on power loss: engage only if already locked; COMFORT stays released	OPEN — LCK-001
REQ-M21	Lateral-g interlock 0.35 / 0.30 g	TARGET (temporary)
REQ-M29	Tower count policy: V-H four towers on both hosts (decided 2026-10-12); six towers remain a module option	BASELINE
REQ-M30	Class S corner mass basis, including distribution allocation, shown by design evidence	OPEN — G1
REQ-M31	Body zoning: only the safety cockpit is suspended; battery, powertrain, fuel, hood and trunk payload on the lower chassis; seam envelope in ICD-001	BASELINE / seam OPEN — ICD-001
REQ-M32	Packaging margin: stack residual ≥ +20 mm at the G1 3-D datum stack, or heights re-baselined by the lever shortfall (table M.9)	OPEN — G1
REQ-M33	Vertical arrest: design vertical impact speed fixed in STR-001 (baseline assumption 3.0 m/s); stroke $s \geq v^2 / (2 \times 196\ m/s^2 \times \eta)$ with $\eta \geq 0.70$ shown by the stop curve; stroke carried in the stack (50 mm placeholder)	OPEN — STR-001 / G1
REQ-M34	Horizontal crash path: loads from lower-body crash structures reach the cell only through the capture frame and links, sized against the 20 g bound of table M.8	OPEN — STR-001 / SAF-001
REQ-M35	Seam function: water, dust, noise and pinch protection; ±57.5 mm normal motion and 107.5 mm overtravel (table M.10); heat, exhaust-gas and fuel-vapour barrier on hybrid hosts (REQ-H07); no battery-gas vent path on electric hosts (REQ-E02)	OPEN — ICD-001 / THERM-001
REQ-M36	Ceiling sizing: design GVW including the Class S exposure and the pack specific-energy sensitivity shall not exceed the ceiling; usable energy follows the rule of table M.7, or the owner re-decides at G1	OPEN — Owner / G1

12. Safety analysis

Screening FMEA exists at L0. No HARA, FTA or independence argument yet. Closure vehicle: SAF-001 (G2). High-severity (S ≥ 9) items require explicit pre-G2 review independent of RPN; RPN scoring does not hide high-severity rows.

ID	Failure mode	S	Action
F-01	Loss of geometric capture	10	Pre-G2 review; redundant paths
F-02	Link / carriage fracture	9	Pre-G2 review; STR-001
F-04	Power-loss valve fails open	9	Fail-closed design; test
F-05	HV fault propagates to module	9	Galvanic isolation; HVIL

13. Verification and gates

Current position L0 (architecture). SIM-001 (G1) and the tower bench (G2) are the first simulation and hardware vehicles. All state-machine transitions require HIL evidence before credit.

Level	Meaning	Typical artefacts
L0	Architecture definition, arithmetic consistency	This baseline
L1	First-order simulation / screening	SIM-001
L2	Component / subsystem bench	Tower bench, ACT-001

Level	Meaning	Typical artefacts
L3	HIL / system integration	State-machine HIL, power-loss timing
L4	Vehicle-level prototype	Instrumented mule
L5	Production-intent / certification	Beyond this baseline

Gate	Focus	Key exit criteria
G0	Document release	Owner release; reviewer appointed
G1	Architecture freeze / packaging	3-D datum stack validating the re-baselined stack with residual ≥ +20 mm or re-baselined heights (REQ-M32); design vertical impact speed fixed (REQ-M33); ICD-001 draft; REG-001; cycle definition; Class S study (REQ-M30); V-L pack rule applied to supplier data (REQ-M36 / REQ-E03)
G2	Subsystem evidence	Tower bench; SAF-001; supplier data; ACT / STR / ENE closure; stop-curve efficiency measured (REQ-M33); horizontal path sized (REQ-M34); seam function, hybrid firewall and electric vent isolation demonstrated (REQ-M35, REQ-H07, REQ-E02)
G3	System HIL	Full state machine; power-loss; lateral-g
G4	Vehicle mule	Instrumented range and structural tests
G5	Production readiness	Beyond this baseline

14. Risks and open items

14.1 Open items (module-level selected)

#	Item	Scope	Closure
3	Core tower P90 limit 67 kg confirmation	Module	G2
5	Lock / S4 logic from COMFORT	Module	ICD-001 / LCK-001
6	State-machine S5–S8 full table	Module	ICD-001
7	Energy buffer sizing, both configurations	Module	ENE-E/H-001
10	High-severity (S ≥ 9) pre-G2 review	Module	SAF-001
11	PARAM-M21-ROLL-GRAD	Module	G2
12	Km confirmation	Module	ACT-001
17	IP filing receipt	Module	Counsel
18	Independent technical reviewer	All	≤2026-12-05 or G1
20	Class S corner mass design evidence and distribution allocation	Module	G1 study (REQ-M30)

Selected module-level items; the full register covering both hosts is Appendix E.

14.2 Selected risks

ID	Risk	L	I	Score	Mitigation
R-22	Re-baselined stack closes (0 / +10 mm) only if all 170 mm of levers are delivered	4	4	16	G1 3-D datum stack
R-M2	P90 module mass exceeds reserve (hybrid lineage R-H2)	4	4	16	G2 supplier correlation
R-M3	Km assumption fails	3	4	12	ACT-001
R-M4	Lock logic ambiguity on power loss	3	4	12	ICD-001 / LCK-001
R-M5	No filing receipt (IP)	3	3	9	Counsel; provisional
R-M6	Class S module +113 kg is covered by the reserves (hybrid 113 kg after D-08, electric 140 kg); design evidence still required	2	3	6	G1 Class S study
R-M7	Height increase raises CdA and mass; range claims move (hybrid V-L cycle ≤70 km/h)	3	3	9	G1 SIM / CFD
R-M8	Electric V-L margin +156 kg (275 kWh) lost if pack specific energy is 200 Wh/kg (−150 kg)	3	4	12	Supplier data; G1 study
R-M9	Seams between the moving cell and fixed bodywork not designed (water, dust, noise, pinch)	3	3	9	ICD-001 / G1
R-M10	Stop-curve efficiency below 0.70 or design impact speed above 3.7 m/s: the 50 mm stroke is not enough and the stack grows	3	4	12	STR-001; Appendix M.5
R-M11	Horizontal crash path into the cockpit unsized (≈58 / 101 kN per link at 20 g)	3	4	12	STR-001 / SAF-001; Appendix M.8

15. Controlled document tree

Document / WP	Role	Owner / gate
MCRS-DOC-00-R8.09	Master — module values (this)	Owner; governs module
MCRS-DOC-00-R5.40	Hybrid host baseline	Owner; hybrid values
MCRS-FE-DOC-00-R7.09	Electric host baseline	Owner; electric values
ICD-001	Interface control document	G1 draft / G2 freeze
SIM-001	System simulation	G1
ENE-E-001 / ENE-H-001	Energy / range / buffer (electric / hybrid)	G1/G2
SAF-001	Safety analysis / HARA	G2
STR-001	Structural / stops	G2
LCK-001	Lock logic and confirmation	G1/G2
ACT-001	Actuator characterisation	G2
REG-001 / REG-FE-001	Regulatory exposure	G1
THERM-001 / THERM-FE-001	Thermal	G2

16. IP, disclosure and commercial paths

Founder / Legal Owner: Amirbahador Atai (individual rights holder). Assignment to a legal entity is a separate controlled act. Licensable module: MCRS-MOD-001. Paths A, B and C as in §1.2. Wording: integration into a two-piece vehicle body (not “EV body”). Trade-mark reference: MCRS (not “MCRS FULL ELECTRIC”). No filing receipt on record; provisional filing under counsel is OPEN. This document is T1 Confidential / NDA controlled; distribution is limited to parties under a valid NDA, and copies outside NDA are uncontrolled. Public disclosure of module architecture or claims beyond the claims table is prohibited until filing and owner release.

17. Claims discipline and glossary

Claim type	Allowed statement	Forbidden statement
Range (electric)	“≥800 km design capability under defined cycle (STUDY)”	“800 km verified / certified range”
Range (hybrid)	“1,000 km target for cycle ≤70 km/h avg (STUDY)”	“1,000 km verified range at any speed”
Energy	“275 kWh (V-L) / 300 kWh (V-H) usable TARGET”	“certified capacity”
Safety	“Invariants defined; screening FMEA at L0”	“ASIL-rated”, “crash-certified”
Comfort	“Targeted isolation (unproven at L0)”	“Superior ride”, clinical outcome
Mass	“P50 closed with reserve; P90 OPEN”	“Mass closed”, “production mass”
Packaging	“FIXED-FOR-ENGAGEMENT / OPEN-P0”	“Locked geometry”, “packaging closed”
Readiness	“TRL 2–3, evidence L0”	“Production ready”, “homologated”
Actuator	“Km ≥63 N/√W ASSUMPTION”	“Proven force capability”

Term	Definition
CMFSB	Captured Mechanical Floating Safety Body — the isolated upper safety cell architecture
MCRS / MCRS-MOD-001	Modular Captured Ride System (product family) / the licensable four- or six-tower module
COMFORT (S2) / SPORT (S2S) / S4	Isolation mode / locked mode with all required locks confirmed / power-loss capture
Class S / M / L	Module corner mass and load class (light / intermediate / heavy host)
P50 / P90 / RSS	Percentile mass estimates; root-sum-square combination of independent deviations
FIXED-FOR-ENGAGEMENT / OPEN-P0	Locked for packaging studies but not closed / blocking packaging item
L0–L5 / G0–G5	Evidence ladder levels / programme gates
BCE / HVIL	MCRS Body Control ECU / High-Voltage Interlock Loop
Usable energy	Net energy available after the reserved / inaccessible fraction (≈10 %)
RA-1	Ambulance / specialty medical transport (comfort context only; no clinical claim)

Part B — Module technical annex

Detailed hydraulic schematics, sensor lists, connector pinouts and CAD envelope live in ICD-001 and supplier-controlled releases. This annex is the single numerical envelope that host baselines cite.

B.1 Stroke and force envelope

Parameter	Value	Status
Macro stroke (body-relative)	±50 mm (100 mm total)	TARGET
Micro stroke	±7.5 mm	TARGET
Force peak / RMS per tower	±2.5 kN / ≈0.7 kN	STUDY
Km	≥63 N/√W at 2–5 mm gap	ASSUMPTION
Control bandwidth / first structural mode	≥20 Hz / ≥60 Hz	TARGET
Mode transition time / design life	≤3 s / ≥10,000 h equivalent	TARGET
Lateral-g inhibit / re-enable	0.35 g / 0.30 g	TARGET (temporary)
Static support floor Class S / L	≥4.5 kN / ≥10.0 kN	DERIVED
Energy buffer	Mandatory for 4- and 6-tower configurations	BASELINE

B.2 Hydraulic parameters (screening)

System pressure, orifice sizing, accumulator volume and fail-safe valve specification are frozen in ICD-001 / STR-001. The passive support path (cylinder to accumulator) remains connected on power loss. The hose-burst static floor is verified under STR-001.

B.3 Sensing and functional independence

Signal	Primary source	Independent / plausibility path
Lateral acceleration	Vehicle dynamics sensor set	Independent channel / cross-check
Yaw and roll rate	Vehicle IMU	Independent channel / model check
Corner position	Dual-channel position sensing where safety-relevant	Cross-corner plausibility
Lock state	Per-lock position / confirmation	Mechanical state correlation
Hydraulic pressure	Pressure transducers	Passive state / rate-of-change plausibility
Battery / HV state	BMS	Independent HV isolation monitor

Redundancy means functional independence, not duplicated identical sensors on one power, processor or signal chain. Common-cause analysis covers shared power, ECU, bus, software, mounting, environment, calibration, vibration and signal processing; a single sensor failure shall not create an unsafe transition.

B.4 Fault classes and recovery

Fault class	Recovery rule
A — transient	S3→S2 only after diagnostic-clear criteria, dwell, sensor plausibility and all interlocks pass
B — latched safety	Authorised reset and recorded review
C — critical capture, structural, lock-integrity or unresolved common-cause	No automatic return to S2 / S2S

B.5 Parameter ID register (module)

PARAM-ID	Description	Value / status	Freeze
PARAM-KM	Lorentz motor constant	≥63 N/√W at 2–5 mm (ASSUMPTION)	ACT-001 G2
PARAM-REF-MASS-MAX	Reference test mass basis	Host-specific (curb + 200 kg)	SIM-001
PARAM-TOWER-MASS-LIMIT	Core tower P90 limit	67 kg conservative until G2	G2 supplier
PARAM-MODE-TIME	COMFORT↔SPORT transition	≤3 s TARGET	HIL G3
PARAM-M21-ROLL-GRAD	Lateral-g filter / gradient	TBD	G2
PARAM-FORCE-PEAK	Peak force per tower	±2.5 kN STUDY	ACT-001
PARAM-STROKE-MACRO / -MICRO	Macro / micro stroke	±50 mm / ±7.5 mm TARGET	ICD-001
PARAM-STATIC-FLOOR-S / -L	Static support floor Class S / L	≥4.5 kN / ≥10.0 kN	STR-001
PARAM-P90-METHOD	P90 combination method (§7.2)	√(n × 223 + 16²) kg deviation; linear-pump sensitivity reported	G2 supplier

B.6 Space claim and interface summary

Tower structure + mounts: 120 mm (Class S planning) / 140 mm (Class L planning). Macro travel + free play: 140 mm. The planning claim is therefore 170 / 190 mm after levers (tower 60 / 80 + travel and play 110) + 50 mm stop and clearance placeholder stroke. After the lever allocation of §9.1 the claim is 170 mm (V-L: tower 60 + travel and play 110) / 190 mm (V-H: tower 80 + 110), plus the 50 mm stop and clearance placeholder. Hardpoint pattern, hydraulic ports, connectors and service access without upper-body removal are defined in ICD-001. The module does not rely on the upper body for primary structural continuity of the lower chassis.

Annex E — Electric reference host (citation of R7.09)

This annex mirrors the electric reference host for interface and mass studies. It is a citation: MCRS-FE-DOC-00-R7.09 controls electric-host values and prevails on any difference. Module values remain under Part A.

E1 Variant definition and packaging

Parameter	V-L	V-H
Architecture	BEV 4×4, four towers (Class S)	BEV 6×6, four towers (Class L)
Length / width	5,000 / 2,000 mm	5,500 / 2,200 mm
Overall height (SPORT)	1,950 mm (re-baselined)	2,130 mm (re-baselined)
Wheelbase	3,000 mm	3,300 mm (front to mid rear tandem)
SPORT ground clearance	200 mm	300 mm
Seats / occupant policy	4 / 400 kg	6 / 600 kg
Battery usable / gross (≈90 %)	275 kWh / ≈306 kWh	300 kWh / ≈333 kWh
Curb target	3,544 kg	5,695 kg
Design GVW	4,044 kg	7,000 kg (ceiling; payload residual 705 kg)
Ceiling / margin	4,200 kg / +156 kg	7,000 kg / 0 kg
Stack residual	0 mm (conditional)	+10 mm (conditional)
Optional protection package	—	≈200 kg; identical treatment to hybrid

E2 Powertrain and HV architecture

Item	V-L	V-H
Nominal HV bus	800 V class	800 V class
Drive topology	Dual e-axle AWD	Front + dual rear e-axles (three)
Front / rear e-motors	≈175–190 kW / ≈175–190 kW	≈150–180 kW / ≈330–370 kW (dual)
Peak / continuous traction	≈350–380 kW / ≈200–220 kW	≈480–550 kW / ≈280–340 kW
AC charger / 0→100 % (V-L 275 / V-H 300 kWh)	22 kW base ≈12.5 h; 44 kW option ≈6.3 h	22 kW base ≈13.6 h; 44 kW option ≈6.8 h
DC fast charge	≥250 kW class (same class as V-H)	≥250 kW class

Pack specific energy of 250 Wh/kg is an aggressive ASSUMPTION; at 200 Wh/kg the V-H 333 kWh gross pack weighs ≈1,665 kg (+333 kg) and the V-L 306 kWh pack ≈1,530 kg (+306 kg). The ≥800 km capability is stated for a defined cycle; the “WLTP-class” label is removed. Fast-charge thermal limits are not yet quantified (THERM-FE-001).

E3 Energy and range screening (V-L 275 kWh / V-H 300 kWh usable)

Case	Wh/km (R7.09)	Range (R7.09)	R7.03 band (superseded)
V-L 80 km/h	≈209	≈1,310 km	≈200–220 Wh/km; ≈1,360–1,500 km
V-L 100 km/h	≈259	≈1,060 km	≈240–260; ≈1,150–1,250 km
V-L 120 km/h	≈321	≈860 km	≈300–330; ≈910–1,000 km
V-H utility 55–60 km/h	≈329–340	≈880–910 km	≈300–340; ≈880–1,000 km
V-H 90 km/h	≈435	≈690 km	≈400–450; ≈670–750 km
V-H 120 km/h	≈579	≈520 km	≈520–580; ≈520–580 km

Inputs: reference mass curb + 200 kg (3,744 kg V-L; 5,895 kg V-H), CdA 0.99 / 1.6 m² (previous 0.86 / 1.5 scaled with the new heights, ASSUMPTION pending CFD), Crr 0.009 / 0.012, battery-to-wheel η 0.88 / 0.87, auxiliary 1 / 2 kW, air density 1.2 kg/m³, no regeneration, grade or temperature credit. V-L keeps ≥800 km up to ≈126 km/h steady speed; V-H keeps ≥800 km only up to ≈72 km/h (utility cycle). The earlier R7.03 bands for V-H utility and V-L 120 km/h were optimistic and are superseded.

E4 Controlled mass budget

Group	V-L (kg)	V-H (kg)
Upper body (CFRP cell + roll + seats + closures)	780	1,250
Battery pack (V-L 275 kWh ≈306 kWh gross; V-H 300 kWh ≈333 kWh gross; at 250 Wh/kg)	1,224	1,332
Chassis, axles, wheels, brakes, steering	820	1,800
e-drive, inverters, HV distribution	180	280
MCRS module (P50; four towers on both hosts)	280	393
HV/LV, cooling, thermal	120	220
Optional protection package	—	200
Reserve (≈4.0 % / ≈3.9 %)	140	220
Curb total	3,544	5,695
Occupants (policy, 100 kg each)	400	600
Payload (trunk; V-L policy, V-H residual to ceiling)	100	705
Design GVW	4,044	7,000
Ceiling / margin	4,200 / +156	7,000 / 0

Check: V-L 780 + 1,224 + 820 + 180 + 280 + 120 + 140 = 3,544; + 400 + 100 = 4,044. V-H 1,250 + 1,332 + 1,800 + 280 + 393 + 220 + 200 + 220 = 5,695; + 600 + 705 = 7,000. The 275 kWh V-L pack saves 108 kg against 300 kWh and gives a +156 kg margin to the 4,200 kg ceiling (+43 kg with the Class S module at 393 kg). At 200 Wh/kg pack specific energy the V-L pack grows by ≈306 kg and the design is 150 kg over the ceiling. The four-tower V-H module (393 kg, down from 565 kg) raises the V-H payload residual from 533 to 705 kg; payload rides in the trunk.

E5 Module mass reference

See §7. Four-tower Class L: P50 393 kg; P90 427 kg (RSS), 439 kg (pump correlated), 565 kg (arithmetic). The 220 kg reserve covers all three cases (+34 / +46 / +172 kg), leaving 48 kg in the arithmetic case; it is a programme reserve shared with every other group. Core tower P90 limit 67 kg until G2.

E6 Structural screening

Case	V-L (upper ≈1,180 kg)	V-H (upper ≈2,050 kg incl. protection)
1 g static, per corner (equal share, ÷4)	≈2.9 kN	≈5.0 kN (≈513 kg, inside the 787.5 kg Class L reference)
2 g vertical	≈5.8 kN	≈10.1 kN
20 g vertical	≈58 kN	≈101 kN
First vertical mode (K-eff 40 kN/m per tower)	≈1.9 Hz	≈1.4 Hz
1 g braking / cornering transfer	TBD — STR-FE-001	TBD — STR-FE-001

Upper sprung mass = cockpit structure + occupants (+ protection package on V-H); payload rides in the trunk and does not load the module (§3.4). Equal-share magnitude screen only; STR-FE-001 supplies the real distribution. Without the protection package the V-H upper mass is ≈1,850 kg. Per-tower stiffness for Class L four-tower is OPEN (ACT / STR-FE-001).

E7 EV safety and host requirements

HVIL, isolation monitoring, precharge, crash isolation, service disconnect and thermal-propagation resistance are host responsibilities defined in R7.09 §6. MCRS safe-state support remains available if traction HV is isolated. No ASIL claim. The battery enclosure does not rely on the upper body for primary containment.

E8 Assumptions, risks and open items (electric)

- Risks (R7.09): R-22 packaging closure conditional on levers (16); R-E3 V-L margin +156 kg lost at 200 Wh/kg pack specific energy (12); R-E6 V-H four-tower 20 g corner load ≈101 kN (8); R-E4 V-H 800 km only on a utility cycle (12); R-E5 regulatory 3,500 kg (9).
- Open: lever feasibility and stack validation, P90 correlation, pack specific energy, cycle definition, lock / S4 logic, energy buffer sizing, stop stroke, Class S mass, height sensitivity of mass and CdA. Full list in R7.09 §17 and Appendix E.

Annex H — Hybrid reference host (citation of R5.40)

This annex mirrors the hybrid reference host. It is a citation: MCRS-DOC-00-R5.40 controls hybrid-host values and prevails on any difference. Module values remain under Part A.

H1 Host definition and powertrain

Parameter	V-L hybrid	V-H hybrid
Architecture	Hybrid 4×4, four towers (Class S)	Hybrid 6×6, four towers (Class L)
Length / width	5,000 / 2,000 mm	5,500 / 2,200 mm
Overall height (SPORT) / ground clearance	1,950 mm / 200 mm	2,130 mm / 300 mm
Wheelbase / seats	3,000 mm / 4	3,300 mm / 6
ICE class / power	V8 4.0 L twin-turbo class / ≈430–450 kW	V12 6.0 L twin-turbo class / ≈460–500 kW
Electric traction / battery	≈150–180 kW / 20 kWh usable	≈200–250 kW / 40 kWh usable
Fuel tank / range target	80 L / 1,000 km (cycle ≤70 km/h avg)	150 L / 700–800 km
Curb / design GVW	2,778 / 3,278 kg	4,983 / 5,983 kg
Ceiling / margin	4,200 / 922 kg	7,000 / 1,017 kg
Stack residual	0 mm (conditional)	+10 mm (conditional)
Optional protection package	—	≈200 kg; identical treatment to electric

V-H tower count is decided: four Class L towers (D-02). The six-tower option (≈99 → 66 kN at 20 g, +≈172 kg module) stays an option of MCRS-MOD-001 and is not baselined.

H2 Packaging stack (re-baselined)

Element (mm)	V-L	V-H	Status
Overall height (SPORT)	1,950	2,130	FIXED-FOR-ENGAGEMENT
Ground clearance (SPORT)	200	300	FIXED-FOR-ENGAGEMENT
Available for structure	1,750	1,830	DERIVED
Chassis / axle structure	280	320	STUDY
Tower structure + mounts (60 mm shared with chassis)	60	80	STUDY
Macro travel (100) + free play (10)	110	110	STUDY
Stop stroke + micro-stage clearance (placeholder)	50	50	ASSUMPTION
Floor-to-H-point	220	220	STUDY
Cabin, H-point to roof	960	960	TARGET
Roof / exterior shell	70	80	STUDY
Allocated sum	1,750	1,820	DERIVED
Residual margin	0	+10	OPEN — P0 (conditional)

Closure uses all 170 mm of levers from the earlier planning table (shared tower depth 60, travel + free play 30, cabin 40, floor 30, shell 10). None is validated; V-L has zero margin. The 50 mm stop-stroke and clearance allowance is a placeholder until STR-001.

H3 Fuel range screening

η base = 0.22 (band 0.18–0.30) until supplier fuel maps (ENE-H-001). Auxiliary load 1 kW (V-L) / 2 kW (V-H); reference mass curb + 200 kg (2,978 / 5,183 kg); CdA 0.99 / 1.6 m² (scaled with the new heights, ASSUMPTION), Crr 0.009 / 0.012; fuel 8.9 kWh/L; battery credit excluded. The 1,000 km target (V-L) is defined for a cycle averaging ≤70 km/h; 90 and 120 km/h are sensitivity only.

Case	Wheel Wh/km	L/100 km (η 0.30 / 0.22 / 0.18)	Range (η 0.30 / 0.22 / 0.18)
V-L 60 km/h	≈119	≈5.1 / 6.9 / 8.5	≈1,580 / 1,160 / 950 km
V-L 90 km/h	≈176	≈7.0 / 9.6 / 11.7	≈1,140 / 840 / 680 km
V-L 120 km/h	≈256	≈9.9 / 13.5 / 16.5	≈810 / 590 / 480 km
V-H 60 km/h	≈244	≈10.4 / 14.1 / 17.3	≈1,450 / 1,060 / 870 km
V-H 90 km/h	≈336	≈13.4 / 18.3 / 22.4	≈1,120 / 820 / 670 km
V-H 120 km/h	≈466	≈18.1 / 24.6 / 30.1	≈830 / 610 / 500 km

At η 0.22 the V-L 1,000 km limit is ≈74 km/h (≈85 km/h at η 0.25), which is why the target cycle is ≤70 km/h; the V-H 700 km target holds to ≈106 km/h and 800 km to ≈92 km/h. V-L does not reach 1,000 km at 120 km/h under any η in the band.

H4 Controlled mass budget

Group	V-L (kg)	V-H (kg)
Upper body (CFRP cell + roll + seats + closures)	755	1,220
Battery (HV pack, 20 / 40 kWh usable class)	145	260
Chassis, axles, wheels, brakes, steering	810	1,780
ICE + hybrid drive + fuel + exhaust (20.9 % / 16.5 % of curb)	580	820
MCRS module (four towers; Class S / L planning, P50)	280	393
HV/LV, cooling, protection (V-H includes optional 200 kg)	95	360
Reserve (4.1 % / 3.0 %; V-L raised by D-08)	113	150
Curb total	2,778	4,983
Occupants (policy) / payload (policy)	400 / 100	600 / 400
Design GVW	3,278	5,983
Ceiling / margin	4,200 / 922	7,000 / 1,017

Check: V-L 755 + 145 + 810 + 580 + 280 + 95 + 113 = 2,778; V-H 1,220 + 260 + 1,780 + 820 + 393 + 360 + 150 = 4,983. Four-corner Class L P90: RSS ≈427 kg, pump-correlated ≈439 kg, arithmetic 565 kg (+34 / +46 / +172 kg versus 393 kg); the 150 kg reserve covers both RSS cases only. The V-L 280 kg module is corner-only (distribution allocation OPEN, §7.3); the V-L reserve was raised from 70 to 113 kg to cover the Class S exposure (D-08). The height increase is assumed mass-neutral (OPEN).

H5 Structural screening loads

Case	V-L (upper ≈1,155 kg)	V-H (upper ≈2,020 kg incl. protection)
1 g static, per corner (equal share)	≈2.8 kN	≈5.0 kN (÷4)
2 g vertical	≈5.7 kN	≈9.9 kN
20 g vertical, four towers	≈57 kN	≈99 kN

Upper sprung mass = cockpit structure + occupants (+ protection package on V-H); payload (trunk) and battery are on the lower chassis (§3.4). Equal-share magnitude screen only; STR-001 supplies the real distribution.

H6 Hybrid safety and host requirements

Fuel, heat and engine excitation near the cell are the principal exposures (THERM-001 / SAF-001). HV isolation invariants are the same as for the electric host. Regulatory: the V-L hybrid design GVW (3,278 kg) is below 3,500 kg; exposure arises from the 4,200 kg ceiling (REG-001).

H7 Assumptions, risks and open items (hybrid)

Primary open items: lever feasibility and stack validation, P90 mass, fuel maps (ENE-H-001), height sensitivity of mass and CdA, lock / S4 logic, regulatory 3,500 kg, energy buffer sizing, stop stroke. Full list in R5.40 §17 and Appendix E.

Appendix A — Correction log

A.1 R8.01 → R8.02 (lineage, retained)

#	Defect (summary)	Action in R8.02
1–3	P90 mass wording (electric / hybrid)	RSS vs arithmetic ranges stated; reserve coverage explicit
4	Energy buffer only six-tower	Mandatory for both 4- and 6-tower
5	Packaging module height	≥260 mm V-L, ≥280 mm V-H
9	Lock vs S4 from COMFORT	Clarified; REQ-M17/M20
10	Document control statements	Master = host-neutral module control; hosts subordinate
11	Figure 1 caption	Six (electric) / four (hybrid baseline, six option)
12	One value one place	Module by citation; host values in host baselines
13	ID inconsistency	Uniform MCRS-MOD-001; PARAM table with IDs
14, 16, 17	REQ-M11 / M29; Master §7 decomposition; ENE reference only E	Present in §11 and §7.2; ENE-E-001 / ENE-H-001 both named
18	Risk R-07	R-H2
19–24	Fuel η, range target, pack density, WLTP, tower limit, equal-share loads	Aligned in annexes and host baselines
28–34	Status tags, R-22, high-S, state machine, regulatory, IP wording	All closed or OPEN with vehicle

A.2 R8.02 → R8.03 (retained)

#	Defect found in R8.02 (Fixed)	Action in R8.03
CL-01	P90 text said the shared pump was added linearly, but 427 / 605 kg reproduce only when the pump is one independent quadrature term (linear gives ≈439 / 618 kg)	§7.2 states the method exactly with RSS, linear-pump and arithmetic cases; PARAM-P90-METHOD added. History: 548 kg was wrong (omitted the 17 kg distribution); 440 kg was reproducible with a linear pump
CL-02	Corrected and Fixed passes shared the same ID and issue date with different content	New ID R8.03, issue date 2026-10-11, §1.3 control rule, Appendix H.2 HA-01
CL-03	Part A claimed no host values outside §9.2 while the annexes held many	Wording corrected: Annexes E/H and Appendices I/K are labelled citations or mirrors, the controlling source prevails, and Appendix D checks them
CL-04	Class table showed corner ≈95 kg beside module 393 kg; A-08 delta (+96 kg) ignored the 17 kg distribution	§7.3 explains 4 × 86 + 32 + 17; Class S exposure +96 to +113 kg; REQ-M30, open item 20, risk R-M6
CL-05	Ceiling tagged LOCKED while V-L electric exceeds it by 152 kg	LOCKED–EXCEEDED rule (Appendix B); §9.2, E1 and open item 21
CL-06	No independent check of host range tables	Annex E3 shows the R7.03 bands beside a Master recomputation; Annex H3 recomputation matches
CL-07	Host range band errors found in R7.03 §7.3	Carried to Appendix H.2 (HA-03)

A.3 R8.03 → R8.04 (retained)

#	Item	Action in R8.04
D-01	Owner decision 2026-10-12: V-L engagement ceiling 4,200 kg (both V-L hosts)	Ceiling updated in §9.2, Annexes E1/H1; electric V-L margin +48 kg; programme policy change recorded under Appendix B
D-02	Owner decision: four Class L towers for V-H on both hosts	Electric V-H module 565 → 393 kg, curb 5,867 → 5,695 kg, payload residual 533 → 705 kg; REQ-M29 BASELINE; six towers remain a module option
D-03	Owner decision: raise SPORT heights to ≈1,950 / ≈2,130 mm	Stack re-baselined (0 / +10 mm, conditional); CdA scaled with height; hybrid V-L cycle ≤70 km/h
D-04	Class S planning value	Kept at 280 kg (target) so the owner ceiling is judged on the design target; exposure disclosed: +113 kg → electric V-L 65 kg over the ceiling, hybrid reserve exceeded
CL-08	Hosts cited superseded Master R8.02	Hosts re-issued as R5.35 and R7.04 citing R8.04 (HA-01 closed)
CL-09	Same 300 kWh pack but DC fast charge ≥150 kW (V-L) vs ≥250 kW (V-H) without rationale	Both ≥250 kW class
CL-10	Host range bands and P90 wording (HA-02, HA-03, HA-04, HA-05)	Corrected in R5.35 / R7.04: method text, recomputed bands, Class S allocation statement, undefined references removed

A.4 R8.04 → R8.05 (retained)

#	Item	Action in R8.05
D-05	Owner decision 2026-10-12: only about half of the overall body (the passenger cell) is suspended; battery and powertrain are in the lower chassis and put no weight on the upper level	New §3.4 (zoning, figure, load boundary, seams, payload location); REQ-M31; open item 23; risk R-M9; zone rules mirrored in both hosts
CL-11	Unclear why battery mass appeared in module discussions	Stated explicitly: the battery loads neither the cell nor the module; it counts only in total vehicle weight against the ceiling and, by thickness, in the chassis stack height
CL-12	Payload was implicitly assumed to ride in the cell	Stated as the conservative baseline; payload in the rear lower zone is a sensitivity (V-H electric upper mass ≈2,050 kg, 20 g corner ≈101 kN, mode ≈1.4 Hz) pending a G1 decision

A.5 R8.05 → R8.06 (retained)

#	Item	Action in R8.06
D-06	Owner decision 2026-10-12: V-L electric battery 275 kWh usable (≈306 kWh gross, 1,224 kg at 250 Wh/kg); V-H electric stays 300 kWh	V-L curb 3,544 kg, GVW 4,044 kg, margin +156 kg (+43 kg with Class S 393 kg; −150 kg at 200 Wh/kg); V-L 800 km holds to ≈126 km/h; AC time 12.5 / 6.3 h; V-H unchanged (800 km only to ≈72 km/h). Reason for keeping V-H at 300 kWh: at 275 kWh its ≥800 km utility range would hold only to ≈63 km/h and the only gain is 108 kg of payload
D-07	Owner decision: only the safety cockpit is suspended; hood and trunk are attached to the lower body; payload rides in the trunk	§3.4 rewritten; upper sprung mass = cockpit + occupants (V-L ≈1,180 kg; V-H ≈2,050 kg with protection); 20 g corner ≈58 / 101 kN (electric), ≈57 / 99 kN (hybrid); first mode ≈1.9 / 1.4 Hz; payload-location open item closed
CL-13	Mass of hood, wings, trunk and rear body panels was not itemised	Assumption A-15: carried inside the chassis group; itemise at the G1 mass freeze (open item 24)
CL-14	Sensitivities in §3.4, E6 and H5 assumed payload in the cabin	Recomputed with payload in the trunk; payload-in-cabin kept as a stated alternative

A.6 R8.06 → R8.07 (retained)

#	Item	Where recorded
EC-01 to EC-10	Consistency defects found in R8.06 (GVW threshold statement, margin value, supersession chain, stop-stroke statements, zone-table battery energy, invariant statement, record order, matrix, IDs, page count)	Appendix M.1
EN-01 to EN-07	Engineering findings (stack margin, stop stroke, horizontal crash path, seams, average coil power, Class S exposure, V-L ceiling sensitivity)	Appendix M.2

A.7 R8.07 → R8.08 (retained)

#	Item	Action in R8.08
D-08	Hybrid V-L reserve raised from 70 to 113 kg to cover the Class S exposure (applied on the owner's instruction to resolve all open defects)	Curb 2,778 kg; design GVW 3,278 kg; margin 922 kg; REQ-H08 BASELINE; R-M6 reduced
RV-01 to RV-12	Review findings of R8.09	Appendix M.3 lists each finding and its resolution; tables M.5 to M.12
Page rule	New page count must be at least the page count of the previous issue	R8.08 has 42 pages against 37 in R8.07

A.8 R8.08 → R8.09 (this issue)

#	Item	Action in R8.09
RX-01 to RX-06	Review findings of R8.09	Appendix M.13; Appendix N
Page rule	New page count at least the previous page count	R8.09 has 45 pages against 42 in R8.08

Appendix B — Status tag definitions (normative)

Tag	Meaning	Change control
BASELINE	Architecture decision frozen for this issue	Requires revision and owner release
TARGET	Design intent under evidence	G1/G2 evidence can refine
STUDY	Screening only; not design commitment	Superseded by supplier / test data
FIXED-FOR-ENGAGEMENT	Locked for packaging studies; not packaging-closed	Re-baseline at G1 if stack fails
DERIVED	Arithmetic consequence of controlled values	Recalculated when inputs change
OPEN — P0	Blocking packaging open item	Must close or re-baseline before G1 exit
ASSUMPTION	Explicit assumption with owner and closure route	Closed by named work package
LOCKED	Hard ceiling or constraint	Programme policy change required
LOCKED–EXCEEDED	A LOCKED value that a controlled design value breaches	Open decision at the named gate; the breach must be shown wherever the value appears

Previous use of LOCKED for open packaging heights was misleading and is replaced by FIXED-FOR-ENGAGEMENT throughout.

Appendix C — Work packages, timeline and release

C.1 Primary work packages

WP	Scope	Gate
SIM-001	System simulation, mode transitions, range sensitivity	G1
ICD-001	Mechanical / electrical / hydraulic / software interfaces; lock protocol; power-loss timing	G1 draft / G2 freeze
ENE-E-001 / ENE-H-001	Energy, range, buffer sizing (electric / hybrid)	G1/G2
SAF-001	HARA, FTA, independence argument, high-S review	G2
STR-001	Stops, structural loads, stop stroke	G2
ACT-001	Actuator force–gap–thermal characterisation	G2
LCK-001	Lock logic, confirmation, power-loss behaviour	G1/G2
REG-001 / REG-FE-001	Regulatory exposure including the 3,500 kg threshold	G1
THERM-001 / THERM-FE-001	Thermal / exhaust / pack heat rejection	G2

C.2 Work package detail

- SIM-001: multi-body and energy simulation of the two-piece architecture under COMFORT and SPORT, power-loss sequences and representative road profiles. Outputs: mode transition timing, force/stroke envelopes, range sensitivity. Gate G1.
- ICD-001: mechanical hardpoints, hydraulic ports, electrical connectors, software interfaces, lock confirmation protocol, power-loss timing (t_det,max, t_valve,max), lateral-g signal path. Draft at G1; freeze at G2.
- ENE-E-001 / ENE-H-001: supplier pack / fuel maps; buffer energy and recharge rate for four-tower and six-tower peak power; charging curves (electric) or fuel maps (hybrid). Outputs: closed range model, buffer specification.
- SAF-001: full HARA, FTA for capture and power-loss, independence argument for the safety monitor, RPN rescoring, explicit pre-G2 review of all severity ≥9 rows. Output: safety case skeleton.
- STR-001: progressive-stop force–displacement, real corner load distribution (CG, payload, transfer), stop stroke for the packaging stack, single-tower fail-safe path.
- ACT-001: Lorentz (or alternative) force–gap–thermal characterisation, Km confirmation or revision, bandwidth and first-mode measurement.
- LCK-001: positive-lock confirmation protocol, incomplete-set inhibit of SPORT, behaviour on power loss from COMFORT vs SPORT, manual rescue sequence. Shared across hosts.

C.3 Indicative timeline alignment

Ph0a: document release (this Master + host baselines). Ph0b (months 1–3): SIM-001 start, ICD-001 draft, reviewer appointment, provisional IP filing. G1 (months 3–5): packaging stack, stack validation, cycle definition, REG-001, Class S study. Ph1 (months 5–10): supplier engagement, bench planning, buffer specification. G2 (months 10–14): tower bench, SAF-001, supplier pack / fuel data. The paid Option (SIM-001 + ICD-001 draft) is a bounded subset of Ph0b/G1 deliverables. Times are indicative and must be re-baselined with owner and supply-chain constraints.

C.4 Gate exit criteria (summary)

Gate	Focus	Key exit criteria
G0	Document release	Owner release; reviewer appointed
G1	Architecture freeze / packaging	3-D datum stack validating the re-baselined stack with residual ≥ +20 mm or re-baselined heights (REQ-M32); design vertical impact speed fixed (REQ-M33); ICD-001 draft; REG-001; cycle definition; Class S study (REQ-M30); V-L pack rule applied to supplier data (REQ-M36 / REQ-E03)
G2	Subsystem evidence	Tower bench; SAF-001; supplier data; ACT / STR / ENE closure; stop-curve efficiency measured (REQ-M33); horizontal path sized (REQ-M34); seam function, hybrid firewall and electric vent isolation demonstrated (REQ-M35, REQ-H07, REQ-E02)
G3	System HIL	Full state machine; power-loss; lateral-g
G4	Vehicle mule	Instrumented range and structural tests
G5	Production readiness	Beyond this baseline

C.5 Release control

This document is T1 Confidential / NDA controlled. Distribution is limited to parties under a valid NDA. Public disclosure of module architecture or claims beyond the claims table is prohibited until filing and owner release. The Independent Technical Reviewer is required no later than the earlier of 2026-12-05 and G1 closure, and before any external safety-performance credit. R8.09 supersedes R8.07 for all Master content. Host baselines R5.40 (hybrid) and R7.09 (electric, 275 kWh V-L / 300 kWh V-H usable) are subordinate to this Master for all module parameters; any conflict on module values is resolved in favour of this Master.

Appendix D — Cross-document consistency matrix

This matrix records, item by item, whether the three controlling documents show the same value as issued on 2026-10-12. “Aligned” means the same value in all three; it does not mean the issue is closed. Open issues are tracked in Appendix E.

Item	Master R8.09	Hybrid R5.40	Electric R7.09	Values aligned?
Module ID	MCRS-MOD-001	MCRS-MOD-001	MCRS-MOD-001	Aligned
Revision identity	R8.09	R5.40 (R5.39 superseded)	R7.09 (R7.08 superseded)	Aligned
Master reference in hosts	—	R8.09	R8.09	Aligned
Body zoning (D-05, D-07)	§3.4: cockpit only suspended; battery, hood, trunk and payload on the lower body	§3 mirror	§3 mirror	Aligned
V-L battery (D-06)	275 kWh V-L / 300 kWh V-H	—	275 / 300 kWh	Aligned
V-H towers	4 both hosts (D-02)	4	4	Aligned
P90 four-tower	427 RSS / 439 linear pump / 565 arith	Same	Same	Aligned
V-L ceiling	4,200 kg (D-01)	4,200 kg	4,200 kg	Aligned
SPORT heights	1,950 / 2,130 mm (D-03)	Same	Same	Aligned
Stack residual	0 / +10 mm conditional	Same	Same	Aligned
Class S module mass	280 kg; exposure +113 kg	280 kg; reserve 113 kg (D-08)	280 kg; reserve 140 kg	Aligned
Hybrid V-L target cycle	≤70 km/h	≤70 km/h	—	Aligned
Electric ranges	Annex E3	—	Same values	Aligned
DC fast charge	≥250 kW class (same class as V-H)	—	≥250 kW both	Aligned
Energy buffer / lock logic / state tags / S≥9	Part A	Same	Same	Aligned
Protection 200 kg	Identical both hosts	Same	Same	Aligned
V-L hybrid reserve (D-08)	113 kg (curb 2,778 kg)	113 kg	—	Aligned
Stop-stroke design speed (RV-01)	3.0 m/s assumption; η 0.70	Same	Same	Aligned
V-L electric pack rule (RV-03)	Table M.7	—	REQ-E03 / table F.4	Aligned
Horizontal bound (RV-06)	Table M.8	Same	Same	Aligned
Gate exit criteria (RX-01)	G1 / G2 include REQ-M30, M32–M36	Same, plus REQ-H07, H08	Same, plus REQ-E02, E03	Aligned
Release readiness (RX-03)	Appendix N.3	Appendix E	Appendix G	Aligned

Appendix E — Full open-item register (module and hosts)

#	Item	Scope	Closure
1	Packaging re-baselined to 0 / +10 mm residual, conditional on 170 mm of unvalidated levers (P0)	Both	G1 3-D datum stack
2	P90 module mass: RSS vs arithmetic; corner correlation	Both	G2 supplier
3	Core tower P90 limit 67 kg confirmation	Module	G2
4	Height sensitivity: mass, CG and CdA effect of +250 / +130 mm (range claims move)	Both	G1 SIM / CFD
5	Lock / S4 logic from COMFORT	Module	ICD-001 / LCK-001
6	State-machine S5–S8 full table	Module	ICD-001
7	Energy buffer sizing, both configurations	Module	ENE-E/H-001
8	Stop stroke and micro-stage clearance in packaging stack	Both	STR-001 / G1
9	Structural corner distribution (CG, transfer)	Both	STR-001
10	High-severity (S ≥ 9) pre-G2 review	Module	SAF-001
11	PARAM-M21-ROLL-GRAD	Module	G2
12	Km confirmation	Module	ACT-001
13	Pack specific energy 250 vs 180–200 Wh/kg	Electric	Supplier
14	Cycle definition for ≥800 km	Electric	SIM / ENE
15	Fuel maps and η confirmation	Hybrid	ENE-H-001
16	Regulatory 3,500 kg threshold (V-L design GVW 3,278 / 4,044 kg; ceiling 4,200 kg)	Both	REG-001
17	IP filing receipt	Module	Counsel
18	Independent technical reviewer	All	≤2026-12-05 or G1
19	Protection package inclusion policy	Both	G1 mass
20	Class S corner mass design evidence and distribution allocation	Module	G1 study (REQ-M30)
21	V-L electric margin +156 kg (275 kWh) holds at 250 Wh/kg pack specific energy; at 200 Wh/kg it is 150 kg over	Electric	G1 study / supplier
22	Lever feasibility: free play 10 mm, 60 mm shared tower depth, 50 mm placeholder for stop stroke	Both	ICD-001 / STR-001 / G1
23	Seam function and envelope between the moving cell and fixed bodywork: ±57.5 mm normal motion, 107.5 mm overtravel, water, dust, noise, pinch; hybrid firewall; electric vent isolation (REQ-M35)	Both	G1 / ICD-001
24	Lower bodywork (hood, wings, trunk, rear body) mass is carried inside the chassis group; itemise at the G1 mass freeze (A-15)	Both	G1 mass
25	Design vertical impact speed (baseline assumption 3.0 m/s) and stop-curve efficiency η ≥ 0.70 (REQ-M33)	Both	STR-001 / G1
26	Horizontal crash path: capture-frame load bound at 20 g (table M.8, REQ-M34)	Both	STR-001 / SAF-001
27	V-L electric pack sizing rule at G1: usable energy ≤ min(275 kWh, table M.7) (REQ-M36, REQ-E03)	Electric	Owner / G1

Appendix F — Claims discipline (full)

No claim of certified range, crashworthiness, homologation, ASIL, production readiness, patentability or freedom to operate is made in this Master or in the host baselines. Evidence level L0, TRL 2–3. OPEN items remain OPEN until evidence closes them.

Claim type	Allowed statement	Forbidden statement
Range (electric)	“≥800 km design capability under defined cycle (STUDY)”	“800 km verified / certified range”
Range (hybrid)	“1,000 km target for cycle ≤70 km/h avg (STUDY)”	“1,000 km verified range at any speed”
Energy	“275 kWh (V-L) / 300 kWh (V-H) usable TARGET”	“certified capacity”
Safety	“Invariants defined; screening FMEA at L0”	“ASIL-rated”, “crash-certified”
Comfort	“Targeted isolation (unproven at L0)”	“Superior ride”, clinical outcome
Mass	“P50 closed with reserve; P90 OPEN”	“Mass closed”, “production mass”
Packaging	“FIXED-FOR-ENGAGEMENT / OPEN-P0”	“Locked geometry”, “packaging closed”
Readiness	“TRL 2–3, evidence L0”	“Production ready”, “homologated”
Actuator	“Km ≥63 N/√W ASSUMPTION”	“Proven force capability”

Appendix G — Glossary (extended)

Term	Definition
CMFSB	Captured Mechanical Floating Safety Body — the isolated upper safety cell architecture
MCRS	Modular Captured Ride System — the product family and licensable module
MCRS-MOD-001	Licensable four- or six-tower isolation module (uniform ID across all documents)
COMFORT (S2)	Isolation mode: locks released, macro + micro active, leveling active
SPORT (S2S)	Locked mode: all required positive locks confirmed, stages held
S4	Power-loss capture: passive support, fail-safe valves, no software dependency for retention
Class S / M / L	Module corner mass / load class (light / intermediate / heavy host)
P50 / P90	50th / 90th percentile mass estimate
RSS	Root-sum-square combination of independent deviations (§7.2 defines the exact method)
FIXED-FOR-ENGAGEMENT	Locked for packaging / engagement studies but not packaging-closed
LOCKED–EXCEEDED	A LOCKED value breached by a controlled design value (Appendix B)
OPEN-P0	Blocking packaging open item
L0–L5 / G0–G5	Evidence ladder levels / programme gates
RA-1	Reference application: ambulance / specialty medical transport (comfort context only; no clinical claim)
BCE	MCRS Body Control ECU
HVIL	High-Voltage Interlock Loop
Usable energy	Net energy available after the reserved / inaccessible fraction (≈10 %)
One value, one place	A value is controlled in one document; any second appearance is a labelled citation or mirror and the controlling source prevails

Appendix H — Document family and supersession

H.1 Document family

Document	Role	Status
MCRS-DOC-00-R8.09	Master — module values (this)	Current
MCRS-DOC-00-R5.40	Hybrid host baseline	Current; subordinate for module values
MCRS-FE-DOC-00-R7.09	Electric host baseline (275 kWh V-L / 300 kWh V-H)	Current; subordinate for module values
MCRS-DOC-00-R8.08 / R8.07 / R8.06 / R8.05 / R8.04 / R8.03 / R8.02 / R8.01	Prior Masters	Superseded
MCRS-DOC-00-R5.39 / R5.38 / R5.37 / R5.36 / R5.35 / R5.34 / R5.33	Prior hybrid baselines	Superseded by R5.40
MCRS-FE-DOC-00-R7.08 / R7.07 / R7.06 / R7.05 / R7.04 / R7.03 / R7.02	Prior electric baselines	Superseded by R7.09
ICD-001 / SIM-001	Interface control / system simulation	G1 draft / pending

Any conflict between host baselines on module parameters is resolved in favour of this Master. Host powertrain, energy and mass values are controlled by the respective host baseline.

H.2 Host re-issue actions (status)

#	Action	Status
HA-01	Re-identify the host baselines and cite the current Master	Closed — R5.40, R7.09 cite R8.09
HA-02	Replace “shared pump linear” with the §7.2 method; add linear-pump sensitivity	Closed — both hosts
HA-03	Correct range bands (electric V-H utility, V-L 120 km/h) and recompute for the new heights	Closed — R7.09 §7.3, R5.40 §7
HA-04	State the distribution allocation in the Class S module mass and the sensitivity	Closed — both hosts
HA-05	Remove undefined CFG-FE-001 and ambiguous R-01; tag ceilings consistently	Closed — R7.09

Appendix I — Safety invariants (mirror of §4)

Restated for readers who use only the annexes. §4 is the controlling statement and prevails on any difference.

Invariant	Rule
Mechanical capture	No operating state permits loss of geometric capture; no free levitation.
Static support	Vehicle weight on passive or fail-safe compliant paths, never the sole magnetic path.
Power loss (S4)	Passive support; powered leveling-supply and hold valves fail-closed; the hydro-pneumatic spring path (cylinder to accumulator) stays connected; software aliveness not required.
Crash path	Progressive stops + structure, independent of closed-loop magnetic force.
Mode transition	SPORT↔COMFORT never enters unconstrained free-float.
SPORT lock fault	Lock failure shall not defeat capture or progressive stops; SPORT is never declared with an incomplete lock set.
Horizontal restraint	Longitudinal / lateral / yaw loads resolved by dedicated paths and/or SPORT lock.
Override limit	Driver override cannot defeat the lateral-g interlock or a critical fault state.
Manual rescue	Two-step rescue available under power loss and hydraulic failure.
HV isolation	Traction HV galvanically isolated from the module control domain; a single HV fault shall not defeat capture or S4.
Thermal / fire	Battery runaway containment is a lower-chassis responsibility and shall not propagate into upper-body capture integrity.
Single-path rule	No single active-control path is credited as the primary crash path.
Lock state on power loss	Locks remain engaged only if already engaged at the instant of loss (SPORT). From COMFORT the locks stay released. Logic frozen in ICD-001 / LCK-001 (REQ-M17 / M20).

Appendix J — Revision and distribution record

Issue R8.09 dated 2026-10-12. Supersedes R8.08. Distributed under T1 NDA only. Copies outside NDA are uncontrolled and must not be used for design or diligence decisions.

Rev	Date	Author	Summary
R8.00	2026-10-08	Atai	First module-centric master with Annex E/H
R8.01	2026-10-08	Atai	Host-neutral wording; Part B; hybrid range; Class L sizing
R8.02	2026-10-09	Atai	Correction-log defects closed; annexes aligned with R5.34 / R7.03 (Corrected and Fixed passes)
R8.03	2026-10-11	Atai	P90 method stated exactly; Class S exposure; LOCKED–EXCEEDED rule; Annex E3 recomputation; host re-issue actions
R8.04	2026-10-12	Atai	Owner decisions D-01 to D-03 (ceiling 4,200 kg; four V-H towers; heights 1,950 / 2,130 mm); hosts re-issued R5.35 / R7.04; range recomputed
R8.05	2026-10-12	Atai	Owner decision D-05: body zoning and load boundary (cell-only suspension, battery in lower chassis, seams, payload location); hosts re-issued R5.36 / R7.05
R8.06	2026-10-12	Atai	Owner decisions D-06 (V-L usable battery 275 kWh; V-H stays 300 kWh) and D-07 (cockpit-only suspension; hood, trunk and payload on the lower body); lower-bodywork mass assumption A-15; sensitivities recomputed; hosts re-issued R5.37 / R7.06
R8.07	2026-10-12	Review	Engineering and consistency review of R8.06: EC-01 to EC-10 and EN-01 to EN-07; REQ-M32 to REQ-M36; hosts R5.38 / R7.07
R8.08	2026-10-12	Atai	Review RV-01 to RV-12 applied; D-08 hybrid V-L reserve 113 kg; stop-stroke efficiency and design speed; pack sizing rule; horizontal capture bound; height, lever and seam sensitivities; hosts R5.39 / R7.08
R8.09	2026-10-12	Atai	Review RX-01 to RX-06 applied: G1 / G2 exit criteria linked to the new requirements; assumption and owner-decision register; release readiness, contract-use statement and release record (Appendix N); hosts R5.40 / R7.09

Approval: TBD after independent technical review. This issue is a proposed baseline pending owner release.

J.1 Post-issue correction record (C-series)

- C-01 (R8.03): P90 combination method stated exactly (pump is an independent quadrature term); linear-pump sensitivity 439 / 618 kg added; PARAM-P90-METHOD created.
- C-02 (R8.03): new revision identity R8.03 and issue date 2026-10-11 replace a second content pass issued under R8.02.
- C-03 (R8.03): Class S module mass clarified as corner-only; exposure +96 to +113 kg; REQ-M30 and R-M6 added.
- C-04 (R8.03): LOCKED–EXCEEDED tag introduced for the V-L electric ceiling breach of that issue; Annex E3 recomputation added.
- C-05 (R8.02): Master aligned to the hosts of that issue (hybrid curb 2,735 / 4,983 kg; electric curb 3,652 / 5,867 kg; superseded by D-02, D-06 and D-08); four-tower Class L P50 = 4 × 86 + 32 + 17 = 393 kg; 6.3 / 9.45 kW identified as peak power.
- C-06 (R8.02): hybrid fuel-range table recomputed with auxiliary load; V-L 1,000 km target cycle average 80 km/h at η 0.22 (superseded: ≤70 km/h since D-03); hybrid regulatory note (design GVW 3,235 kg at that issue, 3,278 kg since D-08).
- C-07 (R8.04): owner decisions D-01 to D-03 applied across Master and both hosts; D-04 keeps Class S at 280 kg with disclosed exposure; hosts re-issued under new IDs.
- C-08 (R8.05): body zoning and load boundary added (§3.4 with side-view figure); battery relationship clarified; payload-location sensitivity added (superseded by D-07 in R8.06); hosts re-issued under new IDs.
- C-09 (R8.06): V-L usable battery 275 kWh (V-H 300 kWh); cockpit-only suspension with hood, trunk and payload on the lower body; loads, modes and masses recomputed; lower-bodywork mass assumption A-15; hosts re-issued under new IDs.
- C-10 (R8.07): engineering and consistency review: EC-01 to EC-10 and EN-01 to EN-07; REQ-M32 to REQ-M36 added; hosts R5.38 / R7.07.
- C-11 (R8.08): review RV-01 to RV-12 applied (stop-stroke efficiency and design speed, D-08 hybrid V-L reserve, pack sizing rule, neutral ceiling table, matrix semantics, horizontal capture bound, height / lever / seam sensitivities, lower-bodywork bound, Class S plan); R8.08 has 42 pages against 37 in R8.07; hosts R5.39 / R7.08.
- C-12 (R8.09): review RX-01 to RX-06 applied: G1 / G2 exit criteria now include REQ-M30 and REQ-M32 to REQ-M36 (and REQ-H07, REQ-H08, REQ-E02, REQ-E03 in the hosts); Appendix N added (assumption and owner-decision register, contract-use statement, release-readiness checklist, release record, remaining work); R8.09 has 45 pages against 42 in R8.08; hosts R5.40 / R7.09.

Appendix K — Quick-reference module envelope

Single-page numerical envelope for MCRS-MOD-001. A mirror of Part A and B.1; Part A controls. All values TARGET / STUDY / ASSUMPTION at L0 unless noted.

Parameter	Value
Towers	4 (Class S); 4 (Class L, both V-H hosts); 6 = module option
Suspended zone	Safety cockpit only (≈½ of body length); battery, powertrain, fuel, hood, trunk and payload on the lower chassis
Macro / micro stroke	±50 mm (100 mm total) / ±7.5 mm
Force peak / RMS per tower	±2.5 kN / ≈0.7 kN
Km	≥63 N/√W at 2–5 mm (ASSUMPTION)
Core tower P50 / P90	48 kg / 67 kg (limit until G2 = 67 kg)
Corner Class S / L P50	≈70 kg / ≈95 kg
Four-tower Class L P50 / P90	393 kg / 427 (RSS), 439 (pump correlated), 565 (arithmetic) kg
Six-tower Class L P50 / P90	565 kg / 605 (RSS), 618 (pump correlated), 815 (arithmetic) kg
Module height claim	170 / 190 mm after levers (tower 60 / 80 + travel and play 110) + 50 mm stop and clearance placeholder
Control bandwidth / first mode	≥20 Hz / ≥60 Hz
Mode time / life	≤3 s / ≥10,000 h
Lateral-g inhibit / re-enable	0.35 g / 0.30 g (temporary)
Supply	48 V class, galvanically isolated from host HV
Peak coil power	≈6.3 kW (4 towers) / ≈9.45 kW (6 towers)
Energy buffer	Mandatory for both 4- and 6-tower configurations
Static floor Class S / L	≥4.5 kN / ≥10.0 kN

Host baselines cite this envelope; they do not redefine it.

Appendix L — Closing statement

This Master Engineering Baseline R8.09, together with the host baselines MCRS-DOC-00-R5.40 (hybrid) and MCRS-FE-DOC-00-R7.09 (electric, 275 kWh V-L / 300 kWh V-H usable), constitutes the set of Document 00 engineering baselines for the MCRS programme as of 2026-10-12.

The numerical, logical, cross-document, assumption and editorial defects identified so far are closed or explicitly carried as OPEN items with named closure vehicles and gates. Module values are controlled exclusively by this Master; host values by the respective host baselines.

Evidence level remains L0 and TRL 2–3. Packaging closure is conditional on unvalidated levers, the electric V-L ceiling margin is +156 kg (+43 kg with the Class S module at 393 kg, negative at 200 Wh/kg pack specific energy), and range figures rest on a CdA scaled with height. No certification, homologation, ASIL assignment, production readiness, patentability or freedom-to-operate claim is made. Independent technical review is required before any external safety-performance credit.

Legal Owner: Amirbahador Atai. Licensable module: MCRS-MOD-001. Classification: T1 Confidential / NDA controlled. Paths A (module licence Option), B (host engagement studies) and C (IP transaction under counsel) remain as defined.

Appendix M — Design and engineering review (R8.09)

This appendix records the review history of the three controlling documents. Review R8.07 (consistency defects EC-01 to EC-10 and engineering findings EN-01 to EN-07) was performed on R8.06; review R8.08 (findings RV-01 to RV-12) was performed on R8.07; review R8.09 (findings RX-01 to RX-06) was performed on R8.08. Calculations are first-order (L0); they set bounds and closure criteria and do not replace evidence from SIM-001, STR-001, ICD-001 or supplier data. Owner decisions D-01 to D-07 are unchanged; D-08 is applied; the status of every assumption is kept in the register of Appendix N.

M.1 Consistency defects closed in R8.07 (retained)

ID	Finding in R8.06	Resolution in R8.07
EC-01	§9.2 stated that every V-L design GVW is above 3,500 kg; the hybrid V-L design GVW is below it.	§9.2 separates electric (above) from hybrid (below; only the 4,200 kg ceiling is above).
EC-02	Appendix L gave an electric V-L ceiling margin of 48 kg (value of R8.04).	Corrected to +156 kg (+43 kg with Class S at 393 kg; −263 kg with Class S at 200 Wh/kg).
EC-03	C.5 said R8.06 supersedes R8.03 and cited R7.06 as 300 kWh; Electric Appendix E said R7.06 supersedes R7.03.	Supersession chain restated; energy stated as 275 kWh V-L / 300 kWh V-H.
EC-04	§10, B.6 and Appendix K stated that stop stroke is not in the stack and that the module claims ≥260 / ≥280 mm.	Stack carries the 50 mm placeholder; B.6 and Appendix K give the planning claim and the re-baselined claim.
EC-05	Zone table listed the electric battery as 300 kWh; V-L electric is 275 kWh.	Corrected in the Master and both hosts; electric mass-table note corrected.
EC-06	§4 said no other section restates the invariants; Appendix I and both hosts mirror them.	§4 is the controlling statement; mirrors are labelled and §4 prevails.
EC-07	Correction records carried superseded values without a label and were not in ID order.	Records in ID order; superseded values labelled.
EC-08	Appendix D reported every item as aligned although EC-01 existed.	Appendix D corrected and extended; in R8.09 the column is renamed (RV-05).
EC-09	New content issued under existing IDs.	Issued under new IDs; family table, matrix, history and supersession updated.
EC-10	Page count must not fall below the original issue.	R8.07 had 37 pages against 35; rule restated for R8.09 (RV-11).

M.2 Design and engineering findings of R8.07 and their status in R8.09

ID	Finding	Resolution / requirement	Status in R8.09
EN-01	The stack closes at 0 mm (V-L) and +10 mm (V-H) with every lever unvalidated.	REQ-M32: residual ≥ +20 mm at G1 or re-baseline the heights (V-L ≈1,970 mm, V-H ≈2,140 mm).	OPEN; consequences quantified in M.9
EN-02	The 50 mm stop stroke has no link to an impact speed.	REQ-M33: STR-001 fixes the design vertical impact speed.	Refined by RV-01 (design speed 3.0 m/s, η 0.70); table M.5
EN-03	The horizontal crash path into the cockpit is not sized.	REQ-M34: loads reach the cell only through the capture frame and links.	Bound in M.8 (RV-06); OPEN STR-001 / SAF-001
EN-04	Seams have a motion requirement only; hybrid front seam faces heat, exhaust gas and fuel vapour.	REQ-M35 (REQ-H07, REQ-E02).	Envelope in M.10 (RV-09); OPEN
EN-05	Average coil power was not stated.	$(700 \div 63)^2 \approx 123$ W per tower: ≈0.49 kW (4 towers), ≈0.74 kW (6 towers), inside 3 / 4 kW.	Closed (calculation)
EN-06	Class S exposure (+113 kg) exceeded the hybrid V-L reserve (70 kg).	Option D-08 (hybrid reserve 113 kg).	Closed by D-08 (RV-02); evidence still required (REQ-M30)
EN-07	Electric V-L loses its ceiling margin at 200 Wh/kg (−150 kg; −263 kg with Class S).	REQ-M36.	Rule in M.7 (RV-03); OPEN owner / G1

M.3 Review R8.08: findings in R8.07 and resolutions

ID	Finding in R8.07	Resolution in R8.08
RV-01	Table M.4 of R8.07 assumed an ideal constant-force stop (η = 1), so 50 mm looked adequate up to 4.4 m/s; a progressive stop is less efficient and the design impact speed was not fixed.	Stop efficiency η = 0.70 and baseline design speed 3.0 m/s (both ASSUMPTION until STR-001): stroke 33 mm against 50 mm; limit of 50 mm is 3.7 m/s (table M.5); REQ-M33 restated.
RV-02	D-08 (hybrid V-L reserve) was left as an unapplied option, so R-H9 stayed open although the hybrid margin is 965 kg.	D-08 applied: reserve 113 kg, curb 2,778 kg, GVW 3,278 kg, margin 922 kg; REQ-H08 BASELINE.
RV-03	Electric V-L sensitivities (−150 kg, −263 kg) had no closure rule.	Pack sizing rule (table M.7): usable energy ≤ min(275 kWh, allowed pack × specific energy × 0.9); 275 kWh holds for ≥241 Wh/kg (Class S at 393 kg) or ≥221 Wh/kg (Class S at 280 kg).
RV-04	Table M.5 of R8.07 labelled the sensitivity “Breaches the ceiling”.	Neutral wording: “Above the ceiling; owner decision at G1” (table M.6).
RV-05	Appendix D used “Aligned” although open issues existed.	Column renamed “Values aligned?” with a legend; open issues stay in Appendix E.
RV-06	The horizontal capture-frame load was not bounded.	First-order bound F = m·a at 20 g and 30 g (table M.8).
RV-07	Height and CdA effects on range were stated only as a percentage.	Quantified for both hosts (table M.9).
RV-08	The consequence of each lever not being delivered was not stated.	Height needed per lever shortfall (table M.9).
RV-09	Seams had only a normal-motion requirement.	Overtravel envelope 107.5 mm and cycle requirements (table M.10).
RV-10	The mass of lower bodywork (A-15) had no bound.	Absorbable-mass bound per host (table M.11).
RV-11	The page rule was “must not fall below the original issue”.	Applied: R8.08 has 42 pages against 37 in R8.07; both hosts also exceed their previous page counts.
RV-12	The Class S study (REQ-M30) had no scope.	Study plan with measurable outputs (table M.12).

M.4 Module requirement additions

ID	Statement	Status
REQ-M32	Packaging margin: stack residual ≥ +20 mm at the G1 3-D datum stack, or heights re-baselined by the lever shortfall (table M.9)	OPEN — G1
REQ-M33	Vertical arrest: design vertical impact speed fixed in STR-001 (baseline assumption 3.0 m/s); stroke $s \geq v^2 / (2 \times 196\ m/s^2 \times \eta)$ with $\eta \geq 0.70$ shown by the stop curve; stroke carried in the stack (50 mm placeholder)	OPEN — STR-001 / G1
REQ-M34	Horizontal crash path: loads from lower-body crash structures reach the cell only through the capture frame and links, sized against the 20 g bound of table M.8	OPEN — STR-001 / SAF-001
REQ-M35	Seam function: water, dust, noise and pinch protection; ±57.5 mm normal motion and 107.5 mm overtravel (table M.10); heat, exhaust-gas and fuel-vapour barrier on hybrid hosts (REQ-H07); no battery-gas vent path on electric hosts (REQ-E02)	OPEN — ICD-001 / THERM-001
REQ-M36	Ceiling sizing: design GVW including the Class S exposure and the pack specific-energy sensitivity shall not exceed the ceiling; usable energy follows the rule of table M.7, or the owner re-decides at G1	OPEN — Owner / G1

M.5 Vertical arrest stroke at 20 g mean deceleration (stop efficiency η = 0.70)

Design impact speed	Drop height $v^2 / 2g$	Stroke, ideal (η = 1)	Stroke at η = 0.70	Consequence
3.0 m/s (baseline assumption)	0.46 m	23 mm	33 mm	Fits the 50 mm placeholder with 17 mm spare
3.7 m/s	0.70 m	35 mm	50 mm	Limit of the 50 mm placeholder
4.4 m/s	1.00 m	50 mm	72 mm	Needs +22 mm in the stack
6.0 m/s	1.83 m	92 mm	131 mm	Needs +81 mm; stack re-baseline

Energy absorbed per corner at 3.0 m/s ($½ \cdot m \cdot v^2$, upper sprung mass ÷ 4): V-L electric 1.33 kJ, V-H electric 2.31 kJ, V-L hybrid 1.30 kJ, V-H hybrid 2.27 kJ. The 3.0 m/s design speed and η = 0.70 are planning assumptions that STR-001 shall confirm or replace; if the confirmed design speed exceeds 3.7 m/s with η = 0.70, the stroke allowance and the stack grow one-for-one (R-M10).

M.6 V-L electric ceiling sizing (4,200 kg; payload 100 kg)

Case	GVW kg	Margin kg	Note
275 kWh, 250 Wh/kg, Class S 280 kg (baseline)	4,044	+156	Baseline
+ Class S module at 393 kg (+113)	4,157	+43	REQ-M30 exposure
+ pack at 200 Wh/kg (+306), Class S 280 kg	4,350	−150	Above the ceiling; owner decision at G1
+ both	4,463	−263	Above the ceiling; owner decision at G1
Option D-09: 225 kWh, 200 Wh/kg, Class S 393 kg (not applied)	4,183	+17	Alternative; superseded by the rule in M.7

M.7 Pack sizing rule for V-L electric (REQ-M36 / REQ-E03)

The V-L design GVW shall not exceed 4,200 kg with payload 100 kg and occupants 400 kg. Everything except the pack and the Class S module is 2,320 kg (3,544 − 1,224). Allowed pack mass is therefore 1,267 kg with the Class S module at 393 kg and 1,380 kg with it at 280 kg. Usable energy ≤ allowed pack mass × gross specific energy × 0.9. The 275 kWh pack (≈306 kWh gross) fits when the specific energy is at least 241 Wh/kg (Class S at 393 kg) or at least 221 Wh/kg (Class S at 280 kg).

Supplier specific energy	Max usable, Class S 393 kg	Adopted usable energy (rule)	Range at 100 km/h / speed holding 800 km	Max usable, Class S 280 kg	Adopted usable energy
250 Wh/kg	285 kWh	275 kWh	≈1,050 km / ≈125 km/h	310 kWh	275 kWh
225 Wh/kg	257 kWh	257 kWh	≈970 km / ≈118 km/h	279 kWh	275 kWh
200 Wh/kg	228 kWh	228 kWh	≈870 km / ≈107 km/h	248 kWh	248 kWh
180 Wh/kg	205 kWh	205 kWh	≈780 km / ≈97 km/h	224 kWh	224 kWh

Rule for G1: with the supplier specific energy and the Class S evidence, usable energy is set to the adopted value of this table (never above 275 kWh); if the owner prefers to keep 275 kWh below the threshold, the ceiling is re-decided instead. The range column uses the worst case (Class S at 393 kg) and the CdA scaled with height. At 180 Wh/kg the adopted energy gives about 780 km at 100 km/h, so the ≥800 km claim then holds only up to ≈97 km/h.

M.8 Horizontal capture-frame load bound (REQ-M34)

Host	Upper sprung mass	Force at 20 g mean	Per link (four links)	Force at 30 g
V-L electric	1,180 kg	232 kN	58 kN	347 kN
V-H electric	2,050 kg	402 kN	101 kN	603 kN
V-L hybrid	1,155 kg	227 kN	57 kN	340 kN
V-H hybrid	2,020 kg	396 kN	99 kN	594 kN

F = m·a on the upper sprung mass (cockpit structure + occupants, no payload, no battery), equal share over four links. The 20 g mean crash pulse is a planning assumption; STR-001 and SAF-001 shall replace it with the cell crash pulse and the real link distribution. Front, rear and side crash structures belong to the lower body; their loads reach the cell only through the capture frame and links.

M.9 Height, CdA and lever sensitivity

Effect of +10 mm height	V-L	V-H
CdA increase	≈0.51 %	≈0.47 %
Electric consumption change (V-L at 100 km/h, V-H at 90 km/h)	+0.29 %	+0.21 %
Electric range change (same points)	−0.29 %	−0.21 %
Hybrid V-L 1,000 km speed limit at η 0.22 (+20 mm shown)	≈74 → ≈73 km/h	—

Lever not delivered	mm lost	Height needed V-L (residual 0)	Height needed V-H (residual 0)
Shared tower depth with the chassis (60 mm)	60	2,010 mm	2,180 mm
Travel + free play reduction (free play 40 instead of 10)	30	1,980 mm	2,150 mm
Cabin H-point to roof 960 → 1,000	40	1,990 mm	2,170 mm
Floor-to-H-point 220 → 250	30	1,980 mm	2,150 mm
Shell thickness 70 / 80 → 80 / 90	10	1,960 mm	2,130 mm
All levers fail	170	2,120 mm	2,290 mm
Margin +20 mm (REQ-M32), all levers delivered	—	1,970 mm	2,140 mm

Baseline heights are 1,950 mm (V-L) and 2,130 mm (V-H) with residual 0 / +10 mm. Every millimetre of lever shortfall adds one millimetre of height and about 0.05 % of CdA; the electric and hybrid range figures move accordingly (first rows).

M.10 Seam envelope (REQ-M35)

Item	Value	Basis
Normal relative vertical motion	±57.5 mm	50 mm macro + 7.5 mm micro
Stop stroke (placeholder)	50 mm	Stack allocation; M.5
Maximum relative travel before the stops bottom	107.5 mm in the compression direction	57.5 + 50
Seal cycling	±57.5 mm over the design life (≥10,000 h equivalent)	Mode and life targets of Part B
One-time overtravel	107.5 mm without seal loss of function	Impact case
Hybrid front seam	Barrier to engine-bay heat, exhaust gas and fuel vapour	REQ-H07
Electric seams	Not a vent path for pack runaway gas	REQ-E02
Pinch protection	Closing gap and force limits frozen in ICD-001	OPEN

M.11 Lower-bodywork mass bound (assumption A-15)

Host	Margin / residual	Lower-bodywork mass that can be absorbed
V-L electric	+156 kg to the 4,200 kg ceiling	up to 156 kg; only 43 kg if the Class S module is also 393 kg
V-L hybrid	922 kg to the 4,200 kg ceiling	up to 922 kg
V-H electric	payload residual 705 kg to the 7,000 kg ceiling	each kg reduces the payload residual
V-H hybrid	1,017 kg to the 7,000 kg ceiling	up to 1,017 kg

Hood, wings, trunk and rear-body panels are budgeted inside the chassis group and are not itemised; the G1 mass freeze shall itemise them (open item 24).

M.12 Class S study plan (REQ-M30)

Output	Target / bound	Why it matters
Class S corner mass	≈70 kg target; 94 kg if equal to Class L	±96 kg on the module
Pump and distribution allocation	32 kg pump + 17 kg distribution	Adds up to 17 kg not in the 280 kg
Static floor basis	≥4.5 kN; occupied share ≈295 kg per corner	Hydraulic and spring sizing
Stop energy	≈1.3 kJ per corner at 3.0 m/s	Stop and stroke design
Horizontal load per link	≈58 kN at 20 g	Capture frame and link sizing

M.13 Review R8.09: findings in R8.08 and resolutions

ID	Finding in R8.08	Resolution in R8.09
RX-01	The G1 and G2 exit criteria did not include the requirements added in R8.07 and R8.08 (REQ-M30, M32 to M36, REQ-H07, H08, REQ-E02, E03), so they were not tied to any gate.	G1 and G2 criteria extended in §13, Appendix C.4 and both hosts (§16).
RX-02	Assumptions and owner decisions were spread over Appendices A, B.5 and M, and the owner's confirmation status was not recorded.	One register (Appendix N.1) with value, basis, owner confirmation and closure gate.
RX-03	The documents did not say what they may be used for.	Contract-use statement (N.2) and release-readiness checklist (N.3).
RX-04	There was no release record.	Release record with open fields for owner release, independent reviewer and counsel (N.4).
RX-05	The path from this baseline to a contract-grade baseline was not listed.	Ordered remaining-work list (N.5).
RX-06	Page rule.	R8.09 has 45 pages against 42 in R8.08.

Appendix N — Assumptions, release readiness and contract use

This appendix collects what a counterparty or counsel needs before relying on the three documents: every assumption and owner decision with its confirmation status, what the documents may and may not be used for, the release conditions, a release record, and the remaining work. “Pending” means the owner has not yet confirmed the item.

N.1 Assumption and owner-decision register

ID	Item	Value	Basis	Owner confirmation	Closure
D-01	V-L engagement ceiling	4,200 kg	Owner decision 2026-10-12	Confirmed	REG-001, G1
D-02	V-H towers	Four Class L on both hosts; six-tower option stays in the module	Owner decision	Confirmed	G1
D-03	SPORT heights	1,950 / 2,130 mm	Owner decision	Confirmed	G1 3-D stack
D-04	Class S planning mass	280 kg target; exposure +113 kg	Owner decision	Confirmed with disclosed exposure	G1 Class S study
D-05 / D-07	Body zoning; cockpit-only suspension; payload in the trunk	Cell ≈ half of the body; hood, trunk, battery, powertrain on the lower chassis	Owner decisions	Confirmed	ICD-001, STR-001
D-06	Usable battery energy	V-L 275 kWh; V-H 300 kWh	Owner decision	Confirmed	Supplier data, G1 / G2
D-08	V-L hybrid reserve	113 kg (curb 2,778 kg)	Applied on the owner's instruction to resolve all defects	Pending	G1 mass freeze
A-01	Stop-curve efficiency	$\eta = 0.70$	Review assumption	Pending	STR-001
A-02	Design vertical impact speed	3.0 m/s	Review assumption	Pending	STR-001
A-03	Crash pulse for the horizontal and vertical bounds	20 g mean	Review assumption	Pending	STR-001, SAF-001
A-04	CdA scaled with height	0.99 / 1.60 m²	Assumption	Pending	CFD, SIM-001
A-05	Height increase mass-neutral	+250 / +130 mm	Assumption	Pending	G1 mass freeze
A-06	Pack specific energy	250 Wh/kg	Supplier assumption	Pending	Supplier data
A-07	Lorentz motor constant Km	≥63 N/√W	Assumption	Pending	ACT-001
A-08	Closure levers	170 mm	Planning	Pending	G1 3-D stack
A-09	Lower bodywork mass inside the chassis group	Not itemised	Assumption A-15	Pending	G1 mass freeze
A-10	Corner mass independence (P90 RSS)	427 / 605 kg RSS	Assumption	Pending	G2 supplier data

N.2 Contract-use statement

The documents may be used for	The documents must not be used for
NDA diligence and technical discussion of the module architecture	Any warranty of range, ride comfort, safety, crashworthiness, homologation, ASIL, mass or packaging
Scoping of the Path A Option (SIM-001 run and ICD-001 draft)	Production, delivery or price commitments
Host engagement studies (interface, packaging, mass) under Path B	Statements to third parties outside the claims-discipline table (Appendix F)
IP-transaction discussion under counsel only (Path C, T3)	Valuation that treats L0 targets as demonstrated performance

N.3 Release-readiness checklist

Condition	Status	Closure vehicle
Owner release of the three documents (G0)	Pending	Owner
Owner confirmation of the register in N.1	Pending	Owner
Independent Technical Reviewer appointed	Not appointed; due no later than the earlier of 2026-12-05 and G1 closure	Owner
Provisional IP filing	No filing receipt on record	Counsel
Counsel review of Path A Option terms and disclosure scope	Open	Counsel
Valid NDA with every recipient	Required per recipient	Owner, counsel
Packaging closure (P0)	Conditional on unvalidated levers	G1 3-D datum stack
SIM-001 first run and ICD-001 draft	Not started (Option deliverables)	Path A

N.4 Release record

Role	Name	Date	Signature
Owner release (Legal Owner)	Amirbahador Atai	To be completed	To be completed
Independent Technical Reviewer	To be appointed	To be completed	To be completed
Counsel review	To be appointed	To be completed	To be completed

The release record is left open on purpose: only the owner, the reviewer and counsel can complete it.

N.5 Remaining work to reach a contract-grade baseline

- Owner confirms or changes every “Pending” item of N.1 and signs the release record.
- Independent Technical Reviewer is appointed and reviews the three documents and the safety invariants.
- Counsel files the provisional application and reviews the Path A Option terms.
- SIM-001 first run and ICD-001 draft (Path A) produce the first non-L0 evidence.
- G1: 3-D datum stack with residual ≥ +20 mm, Class S study, stop-curve and impact-speed definition, pack rule applied to supplier data.
- G2: tower bench, SAF-001, horizontal path, seam function, supplier mass and correlation data.

— END OF MCRS-DOC-00-R8.09 —